July 25, 2025

Office of Food Additive Safety (HFS-200)
Center for Food Safety and Applied Nutrition
Food and Drug Administration
5001 Campus Drive
College Park, MD 20740

Subject: Submission of a Generally Recognized as Safe (GRAS) Notification for Farmtory™ Brazzein

Dear Sir or Madam,

On behalf of MicroFarmtory LLC, we are pleased to submit this GRAS notification in accordance with 21 CFR §170.203-§170.285 to inform the U.S. Food and Drug Administration (FDA) of the determination that Farmtory™ Brazzein is Generally Recognized as Safe (GRAS)for its intended use.

Farmtory™ Brazzein is manufactured using a genetically engineered strain of *Komagataella phaffii* and contains ≥90% brazzein. It is intended for use as a general-purpose sweetener in food, excluding infant formula, meat, and poultry products.

The GRAS determination has been made by a qualified panel of independent scientific experts, who have critically evaluated the available scientific data, safety assessments, and intended use conditions. Based on a comprehensive review, the GRAS Panel has unanimously concluded that the intended use of Farmtory™ Brazzein meets the standard of reasonable certainty of no harm and does not require premarket approval under the Federal Food, Drug, and Cosmetic Act.

We appreciate the FDA's review of this notification and look forward to your acknowledgment of the submission. Please do not hesitate to contact us if additional information or clarification is required.

Thank you for your time and consideration.

Sincerely,

Wenwen Ma, Ph.D.
PureNutrix+ Inc
400 West Capitol Avenue, Suite 1700
Little Rock, AR, 72201
Tel: 501-800-4108
Email: regulatory@purenutrix.us

GRAS NOTICE FOR A SWEET PROTEIN FARMTORY™ BRAZZEIN

SUBMITTED TO:
Office of Food Additive Safety (HFS-200)
Center for Food Safety and Applied Nutrition (CFSAN)
Food and Drug Administration
5001 Campus Drive College Park, MD
20740 USA

SUBMITTED BY:
MicroFarmtory LLC,
163 Capitol Ave, Suite 413, C1088
Cheyenne, WY 82001

PREPARED BY:
PureNutrix+ Inc. (Agent)
400 West Capitol Avenue, Suite 1700
Little Rock, AR 72201
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DATE: 07/22/2025

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PART 1 SIGNED STATEMENTS AND CERTIFICATION

In accordance with 21 CFR §170 Subpart E, specifically §170.203 through §170.285, MicroFarmtory LLC (MicroFarmtory) hereby notifies the United States (U.S.) Food and Drug Administration (FDA) of the intended uses of Farmtory™ Brazzein, a product equivalent to native brazzein extracted from the Oubli Fruit, derived from *Komagataella phaffii*. Farmtory™ Brazzein, containing ≥ 90% brazzein, is manufactured for MicroFarmtory. The intended use of Farmtory™ Brazzein as a general-purpose sweetener in food, excluding infant formula, meat, and poultry products, is not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act. This determination is based on our review that the notified uses of Farmtory™ Brazzein are Generally Recognized as Safe (GRAS). Furthermore, as a responsible official of MicroFarmtory, the undersigned certifies that all data and information presented in this notice represent a complete, accurate, and balanced submission. We confirm that both favorable and unfavorable information known to MicroFarmtory has been considered and is pertinent to the evaluation of the safety and GRAS status of Farmtory™ Brazzein as a food ingredient for use in conventional food and beverage products, as outlined herein.

Signed,

[redacted]

Oliver Yu, Ph.D.
Title: Chief Science Officer
MicroFarmtory LLC
Email: oliver.yu@microfarmtory.com

Date: 2025.7.25

1.1 Name and Address of Notifier and Agent of the Notifier

Notifier

MicroFarmtory LLC,
163 Capitol Ave, Suite 413, C1088
Cheyenne, WY 82001
Manufacturer: AquaFarmtory Biotechnology Ltd.,
68 Xinqing Road, Suite 1-1105, SIP, Jiangsu, PRC.

Agent of the Notifier

PureNutrix+ Inc.
400 West Capitol Avenue, Suite 1700
Little Rock, AR 72201
USA
Attn: Wenwen Ma, Ph.D.
Email: regulatory@purenutrix.us
Tel: 501-800-4108

1.2 Common Name

Farmtory^{TM} Brazzein

1.3 Conditions of Use

Farmtory^{TM} Brazzein, produced by MicroFarmtory, is intended for use as a general-purpose sweetener in food in accordance with current Good Manufacturing Practice (cGMP), excluding infant formula and meat and poultry items.

1.4 Basis for GRAS

Pursuant to 21 CFR § 170.30 (a)(b) of the Code of Federal Regulations (CFR) (U.S. FDA, 2022a), we conclude that the intended uses of Farmtory^{TM} Brazzein as described herein are GRAS based on scientific procedures.

1.5 Availability of Information

The data and information that serve as the basis for this GRAS Notification will be sent to the U.S. FDA upon request or will be available for review and copying at reasonable times at the offices of:

PureNutrix+ Inc. (PureNutrix)
400 West Capitol Avenue, Suite 1700
Little Rock, AR 72201 USA
Attn: Wenwen Ma, Ph.D.
Email: regulatory@purenutrix.us
Tel: 501-800-4108

Should the U.S. FDA have questions or additional information requests regarding this GRAS Notice, PureNutrix will supply these data and information upon request.

1.6 Freedom of Information Act, 5 U.S.C. 552

It is MicroFarmtory's view that all data and information presented in Parts 2 through 7 of this Notice do not contain trade secret, commercial, or financial information that is privileged or confidential, and therefore, all data and information presented herein are not exempted from the Freedom of Information Act, 5 U.S.C. 552.

PART 2 IDENTITY, METHOD OF MANUFACTURE, SPECIFICATIONS, AND PHYSICAL OR TECHNICAL EFFECT

2.1 Identity

2.1.1 Identity of Farmtory™ Brazzein

Brazzein is a sweet-tasting protein found in the fruit of the native West African Oubli plant (*Pentadiplandra brazzeana*) that has a sweetness approximately 500-2000 times that of sucrose (Ming & Hellekant, 1994). The natural brazzein has two isoforms: a major isoform with 54 amino acids (brazzein-54, representing about 80% of brazzein in the Oubli fruit, or ~80%) and a minor but sweeter isoform with 53 amino acids (brazzein-53, ~20%) (Neiers et al., 2021).

Farmtory™ Brazzein, the subject of this GRAS notice, is produced using a strain of *Komagataella phaffii (K. phaffii)* (also known as *Pichia pastoris*) that has been genetically engineered to express brazzein-53 protein via fermentation. The final product is comprised of ≥ 90% brazzein-53 protein, with minor constituents such as residual host cell proteins carried over from the fermentation process of *K. phaffii*, ash, moisture, carbohydrates, and fat.

The amino acid sequence of Brazzein-53 in Farmtory™ Brazzein is as follows, the same as the native Brazzein-53.

DKCKKVYENYPVSKCQLANQCNYDCKLDKHARSGECFYDEKRNLQCICDYCEY

2.1.2 Proteomics Analysis of Farmtory™ Brazzein

The identity of the brazzein constituent of Farmtory™ Brazzein has been demonstrated via proteomics analysis. The result shows a high degree of sequence coverage 96.2% with 12 peptides matching the amino acid sequence to the native brazzein-53 (Figure 1).

Brazzein-53	D K C K K V Y E N Y P V S K C Q L A N Q C N Y D C K L D K H A R S G E C F Y D E K R N L Q C I C D Y C E Y	
Mapped Peptides	C K K V Y E N Y P V S K	S G E C F Y D E K R
	K V Y E N Y P V S K	S G E C F Y D E K R N L Q C I C D Y C E Y
	V Y E N Y P V S K C Q L A N Q C N Y D C K L D K	N L Q C I C D Y C E Y
	V Y E N Y P V S K C Q L A N Q C N Y D C K	
	V Y E N Y P V S K C Q L A N Q C N Y D C K L D K H A R	
	C Q L A N Q C N Y D C K L D K	
	C Q L A N Q C N Y D C K	

Figure 1 Sequence Coverage from Peptide Mapping of Brazzein-53 in Farmtory™ Brazzein

In addition, the product has been analyzed with UV and high performance liquid chromatography (HPLC) against the reference standard. Taken together, it is confirmed that the brazzein in Farmtory™ Brazzein is chemically identical to the native brazzein isolated from Oubli fruit.

2.1.3 Protein Structure of Farmtory™ Brazzein

Brazzein structures can be predicted based on the amino acid sequence using public software such as AlphaFold3 (Abramson et al., 2024). The predicted structure can be compared to the stored structure from X-ray crystallography in the public database (Caldwell et al., 1998). An AlphaFold3 structure prediction was performed by using the amino acid sequence deduced from the LC-MS/MS analysis. Figure 3 below demonstrates that the 3D protein structure of Farmtory™ Brazzein (A) matches the structure of brazzein (B) in the protein database PDBe-KB (P56552) by X-ray crystallography.

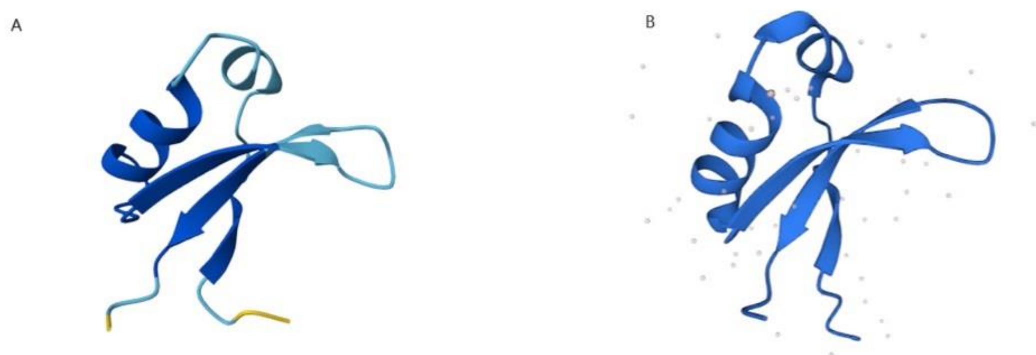

Figure 2 Brazzein Structure

(A) Structure of Farmtory™ Brazzein from the Google Protein database (2025-05-27).

(B) Brazzein structure from Protein Database PDBe-KB (P56552).[1]

These data indicate that brazzein-53 in Farmtory™ Brazzein is substantially equivalent to native brazzein.

2.2 Method of Manufacture

Due to the geographic restrictions of Oubli plants and the low production yield of brazzein, different genetic engineering methods have been developed for the biosynthesis of brazzein for human consumption. Farmtory™ Brazzein is produced by fermentation using *K. phaffii*, a genetically engineered organism, following a standard industrial fermentation and purification process that complies with current Good Manufacturing Practice (cGMP) and Hazard Analysis and Critical Control Points (HACCP) principles, and meets appropriate food grade specifications.

2.2.1 Production Microorganism

2.2.1.1 Parental (Host) Organism

The parental (host) organism of Farmtory™ Brazzein is *Komagataella phaffii. (K. phaffii)* (formerly referred to as *Pichia pastoris* or *P. pastoris*), a yeast species widely used in the food

1 www.ebi.ac.uk/pdbe/pdbe-kb/proteins/P56552

and pharmaceutical industries. *K. phaffii* is classified as a Biosafety Level 1 (BSL-1) organism and has qualified presumption of safety (QPS) status in the EU for use in enzyme production (EFSA, 2017). It is a well-characterized, non-toxigenic, and non-pathogenic microorganism with an established history of safe use in food production. Its safety has been acknowledged in multiple GRAS Notices submitted to the U.S. FDA, including GRN Nos. 737, 967, 1001, 1056, 1104, 1142, and 1167.

The taxonomic identity of *K. phaffii* is presented in Table 1. below.

Table 1 Taxonomic Identity of *Komagataella phaffii*

Phylum	Ascomycota
Class	Saccharomycetes
Order	Saccharomycetales
Family	Phaffomycetaceae
Genus	*Komagataella*
Species	*phaffii*

Farmtory^{TM} Brazzein is produced by precision fermentation using a strain of *K. phaffii* genetically engineered to express genes encoding for the biosynthesis of brazzein. The production strain is derived from the well-characterized host organism *K. phaffii* NRRL Y11430. The safe strain lineage of *K. phaffii* NRRL Y11430 is largely based on the recognized nonpathogenic and nontoxigenic nature of this strain, as well as the fact that the species itself has not been associated with any known adverse effects throughout its extensive history of use in food production. Furthermore, proteins from strains derived from *K. phaffii* NRRL Y11430 have been demonstrated to lack allergenic and toxigenic potential (Jin et al., 2018; Reyes et al., 2021). According to criteria established by the Joint FAO/WHO Expert Committee on Food Additives (JECFA), *K. phaffii* NRRL Y11430 is considered a safe strain lineage suitable for use as a host organism in food production (FAO/WHO, 2020).

The host organism, *K. phaffii* GS115, has been demonstrated to be genomically similar to *K. phaffii* NRRL Y11430 and does not contain any native plasmids or antibiotic resistance genes.

2.2.1.2 Production Host Strain *K. phaffii* GS115

2.2.1.2.1 Origin of production strains

K. phaffii GS115 (formerly *P. pastoris* GS115), the production strain used for Farmtory^{TM} Brazzein, is a derivative of *K. phaffii* Y-11430 by mutagenesis. Extensive reports have been published to compare these two strains, with detailed genome comparisons, including genome sequence and gene expression analysis, which are publicly available (Brady et al., 2020).

2.2.1.2.2 Construction of production strain

The following steps were taken through targeted manipulations to generate a brazzein production strain in the host yeast *K. phaffii*. First, a set of protease genes was removed from the yeast genome by CRISPR technology to reduce protein degradation and increase yield. Second, a set of chaperones and protein processing assistant genes was over-expressed in the genome. These genes help the proper folding of the brazzein protein and increase the secretion of the final product. All genes used in this step are original *K. phaffii* genes. Finally, the brazzein coding gene was over-expressed in the host. The brazzein gene was codon optimized using the commercial pHKA vector. The promoter and terminator in the expression cassette are native *K. phaffii* sequences. The gene encoding brazzein is based on the original plant gene from *Pentadiplandra brazzeana* (UniProtKB P56552). The brazzein expression cassette was stably integrated into the genome for selection. Importantly, the production strain has no residual plasmids or antibiotic resistance genes. Thus, the transfer of plasmids or antibiotic resistance genes to unrelated microbes is not expected. The final product is not expected to contain antibiotics because no antibiotics are used during the production process. The strain has been deposited in a public database: China Center for Type Culture Collection (CCTCC).

2.2.2 Description of the Production Process

Farmtory^{TM} Brazzein is made by precision fermentation, following a typical fermentation protocol. As shown by the process flow chart in Figure 4, the glycerol stock of the engineered strain is first propagated through multiple seed cultures. These cultures are carefully monitored for cell growth and other critical parameters before being used to inoculate the main fermentation vessel. During the main fermentation, the yeast cells are cultured under controlled conditions, with continuous monitoring and regulation of key process variables such as optical density (OD), wet cell weight, pH, and temperature.

After fermentation, the yeast culture broth is processed to purify brazzein. The supernatant fraction from the yeast culture is heated to kill the production host microorganism. After cooling and centrifugation, the supernatant fraction is collected and passed through ultrafiltration membrane to remove macromolecular proteins. The supernatant fraction is then loaded onto an ion exchange resin column for purification. Following washing and elution with a gradient concentration of NaCl solution, the eluates containing Farmtory^{TM} Brazzein are collected, desalted by nanofiltrations to remove NaCl. After drying, the resulting Farmtory^{TM} Brazzein is ground, sifted, and packed in food grade sealed double PE bags that are placed in aluminum alloy drums.

All raw materials, processing aids, filtration agents, and packaging materials in the manufacturing process are safe and suitable standard ingredients, are used in in accordance with appropriate U.S. regulations, conform to either specifications outlined in the Food Chemicals Codex (FCC) or to other applicable regulatory standards, are GRAS for their intended uses, or are subject of an effective food contact notification. The final product is a dried powder packed in tightly sealed food grade double PE bags that are kept in the aluminum alloy drums.

The manufacturing process for Farmtory™ Brazzein complies with current Good Manufacturing Practice (cGMP) and Hazard Analysis and Critical Control Points (HACCP) as shown in the flow chart figure 3.

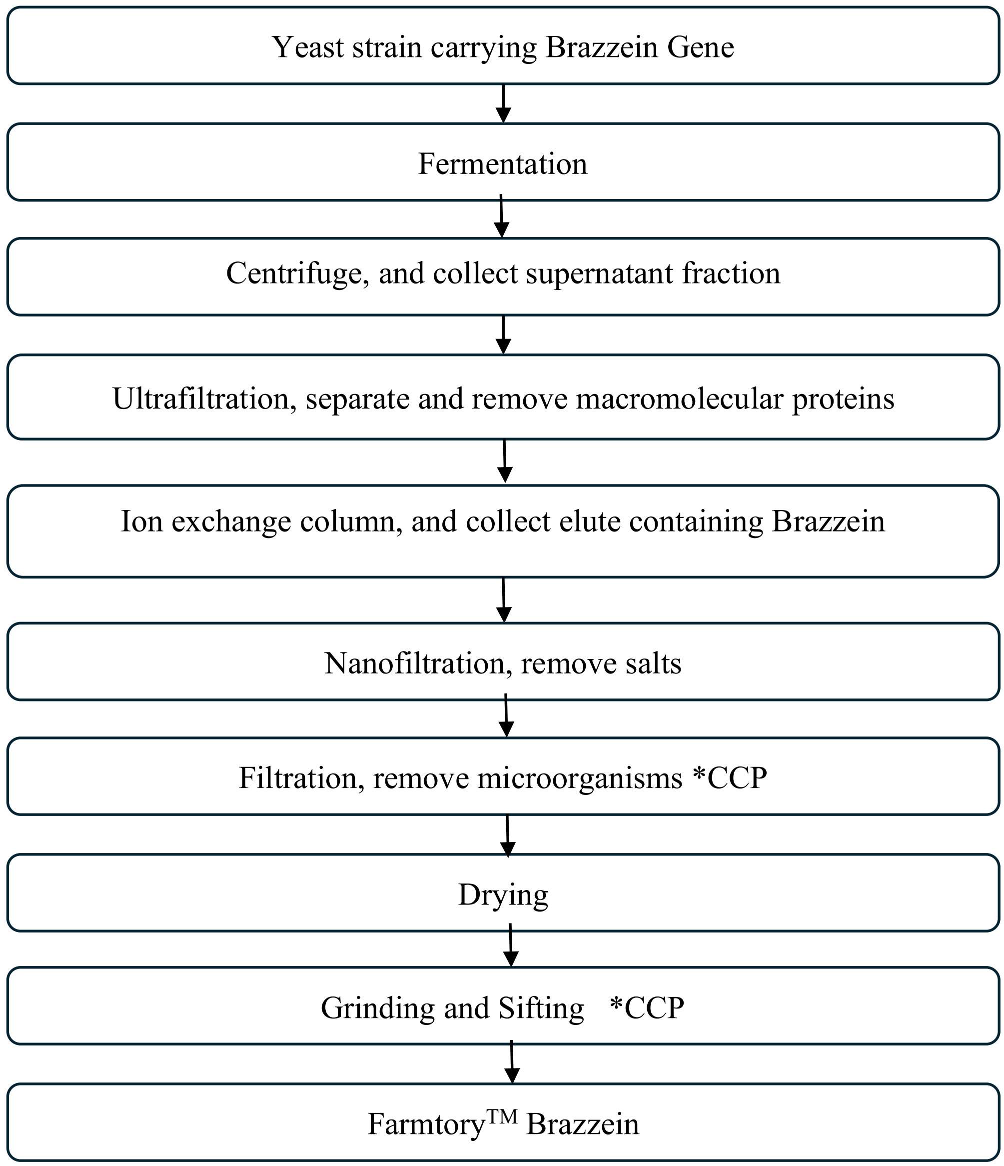

Figure 3 Flow Chart of Manufacturing Process for Farmtory™ Brazzein

*CCP= Critical Control Point

2.3 Product Specifications and Batch Analyses

2.3.1 Specification

The specifications of Farmtory™ Brazzein are shown in Table 2. The attributes include total protein, fat, ash, and carbohydrates, moisture (loss on dry) as well as the brazzein content, which is calculated as the percentage of total mass, as well as the percentage of the total protein mass in the sample. Additionally, specifications have been established for heavy metals (arsenic, cadmium, lead, mercury) and microbiological limits. Each parameter is measured using internationally recognized methods, such as those from the Association of Official Analytical Collaboration (AOAC). The brazzein content is measured using HPLC against the external brazzein reference standard.

Table 2 Specifications for Farmtory™ Brazzein

Parameter	Specification	Method
Appearance	Yellowish brown powder	GB/T 10220-2012
Odor	Characteristic	GB/T 10220-2012
Taste	Intense Sweetness	GB/T 10220-2012
Total protein (% w/w)	≥ 90%	AOAC 930.29
Brazzein (% w/w)	≥ 90%	HPLC (QC-00-01)
Loss on Drying	≤ 10%	AOAC 930.04
Fat %	≤ 1%	AOAC 996.06
Ash (% w/w)	≤ 5%	AOAC 930.30
Carbohydrates	≤ 5%	FDA21 CFR101.9
Arsenic	≤ 0.2 ppm	AOAC 2015.01 ICP-MS
Lead	≤ 0.1 ppm	AOAC 2015.01 ICP-MS
Cadmium	≤ 0.1 ppm	AOAC 2015.01 ICP-MS
Mercury	≤ 0.1 ppm	AOAC 2015.01 ICP-MS
Total plate count	≤ 5,000 CFU/g	AOAC 990.12
Molds and Yeasts	≤ 100 CFU/g	AOAC 997.02

Total Coliform	≤ 10 CFU/g	AOAC 991.14
E. Coli	≤ 10 CFU/g	AOAC 991.14
Salmonella spp.	Not detected in 25g	AOAC 967.26
Listeria	Not detected in 25g	AOAC 993.12

GB/T: National Standard Recommended of China; AOAC: Association of Official Analytical Collaboration; HPLC: High Performance Liquid Chromatography; ICP-MS: Inductively Coupled Plasma Mass Spectrometry; CFU: Colony Forming Unit.

2.3.2 Batch Analysis

Table 3 summarizes the analysis of three non-consecutive production batches of Farmtory^{TM} Brazzein. The batch data confirm production consistency and adherence to the specifications outlined in Table 2.

Table 3 Analytical Data of Three Non-Consecutive Manufacture Batches

Parameter	**Specification**	Batch 20240730	Batch 20240820	Batch 20240929
Appearance	Yellowish brown powder	Yellowish brown powder	Yellowish brown powder	Yellowish brown powder
Odor	Characteristic	Characteristic	Characteristic	Characteristic
Taste	Intense Sweetness	Intense Sweetness	Intense Sweetness	Intense Sweetness
Total protein (% w/w)	≥ 90%	91.80%	92.50%	91.30%
Brazzein (% w/w)	≥ 90%	91.00%	91.5%	90.00%
Loss on Drying	≤ 10%	5.00%	5.2%	5.40%
Fat %	≤ 1%	ND	ND	ND
Ash (% w/w)	≤ 5%	1.20%	1.5%	0.82%
Carbohydrates	≤ 5%	2.00%	0.80%	2.48%
Arsenic	≤ 0.2 ppm	ND	ND	ND
Lead	≤ 0.1 ppm	ND	ND	ND
Cadmium	≤ 0.1 ppm	ND	ND	ND
Mercury	≤ 0.1 ppm	ND	ND	ND
Total plate count	≤ 5,000 CFU/g	550 CFU/g	720 CFU/g	640 CFU/g

Molds and Yeasts	≤ 100 CFU/g	<10 CFU/g	<10 CFU/g	<10 CFU/g
Total Coliform	≤ 10 CFU/g	< 10 CFU/g	< 10 CFU/g	< 10 CFU/g
E. coli	≤ 10 CFU/g	< 10 CFU/g	< 10 CFU/g	< 10 CFU/g
Salmonella spp.	Not detected in 25g	ND	ND	ND
Listeria	Not detected in 25g	ND	ND	ND

ND, not detected; CFU, colony forming units; Fat limit of detection (LOD) = 0.01 %. Arsenic limit of quantitation (LOQ) = 0.002 mg/kg; Lead LOQ = 0.02 mg/kg; Cadmium LOQ = 0.002 mg/kg; Mercury LOQ = 0.001 mg/kg.

In addition, absence of transgenic DNA in multiple Farmtory™ Brazzein samples was determined using methods based on the guidelines provided by the European Food Safety Authority (EFSA, 2011). The PCR analysis confirmed that no detectable transgenic DNA is present in Farmtory™ Brazzein at levels above the detection limit of <0.01 ng/g (Protection, 2011).

2.3.3 Mineral Profile

The mineral profile of Farmtory™ Brazzein was measured, and the result is provided in Table 4.

Table 4 Trace Elements in Farmtory™ Brazzein

Element	Method	Result (ppm or mg/kg)
Phosphorus（P）	AOAC 984.27	28.83
Calcium（Ca）	AOAC 984.27	32.2
Copper（Cu）	AOAC 984.27	3.9
Iron (Fe)	AOAC 984.27	17.8
Potassium（K）	AOAC 984.27	2.4
Magnesium（Mg）	AOAC 984.27	6.1
Manganese（Mn）	AOAC 984.27	0.3
Sodium（Na）	AOAC 984.27	4011.4
Zinc （Zn）	AOAC 984.27	ND

ND: Not detected. AOAC: Association of Official Analytical Collaboration; Zinc limit of quantitation (LOQ) = 1 ppm.

PART 3 DIETARY EXPOSURE

3.1 Intended Use and Estimated Intake of Farmtory™ Brazzein

Farmtory™ Brazzein is intended for use as a general-purpose sweetener in foods and beverages in accordance with cGMP, excluding infant formula and meat and poultry products. To estimate dietary exposure to Farmtory™ Brazzein, the Renwick method (Renwick, 2008), a well-established model for estimating intake of high-intensity sweeteners, was applied.

Renwick (2008) used the available post-market surveillance data for other high-intensity sweeteners (i.e., aspartame, steviol glycosides) as the basis for the assessment of dietary exposure for rebaudioside A (reb A), a commonly consumed steviol glycoside, by assuming full replacement of the currently approved intense sweeteners with the new sweetener. This assessment methodology yields intake estimates that, while conservative, are realistic in that they reflect actual post-market intakes of high-intensity sweeteners. Specifically, to estimate reb A intakes, Renwick calculated the post-market surveillance intake estimates for intense sweeteners presently used in the global marketplace as sucrose equivalents in various population groups (Table 5). The data used in these analyses were primarily derived from studies that used specifically designed food diaries combined with actual use-levels or approved levels in different foods and beverages. In order to predict dietary exposure to reb A, the intake estimates for the high-intensity sweeteners were adjusted for the sweetness intensity of reb A relative to sucrose, which is approximately 250.

The majority of other high-intensity sweeteners (aspartame, steviol glycosides, etc.) have been approved by the FDA as general-purpose sweeteners without their uses being restricted to specific foods or use-levels. The foods to which high-intensity sweeteners are added and their use-levels are dominated by technological properties such as sweetness potency. The Renwick method has been applied in various GRAS notices on sweeteners that FDA reviewed and issued no questions letters (i.e., recent GRN 1106, GRN 1142, GRN 1167)

Brazzein is 500–2000 times sweeter than sucrose, depending on the concentration of the comparison solution, as reported in the public literature (Ming & Hellekant, 1994). We conservatively assume that Farmtory™ Brazzein is 500 times as sweet as sucrose and apply the Renwick method (Renwick, 2008) to estimate the intake of brazzein. The resulting intake estimates of Farmtory™ Brazzein for its intended uses as a general-purpose sweetener, expressed in mg/kg body weight per day, are presented in Table 5 for both average and high (≥ 90th percentile) consumers across different subpopulations.

Table 5 Estimated Daily Intake of Farmtory™ Brazzein Using the Renwick Method

	Intake of Intense Sweeteners (Expressed as Sucrose Equivalents) (mg/kg bw/day)[a]	Estimated Intake of Farmtory™ Brazzein (mg/kg bw/day)[b]

Population Group	Average Consumer	"High" Consumer ≥90th Percentile	Average Consumer	"High" Consumer ≥90th Percentile
Non-Diabetic Adults	255	675	0.51	1.35
Diabetic Adults	280	897	0.56	1.79
Non-Diabetic Children	425	990	0.85	1.98
Diabetic Children	672	908	1.34	1.82

[a] Estimates as reported by Renwick (2008).

[b] Calculated assuming Farmtory™ Brazzein is approximately 500 times as sweet as sucrose.

As shown in Table 5, the highest estimated intake of Farmtory™ Brazzein is 1.98 mg/kg bw/day for non-diabetic children at the high-end (or ≥90th percentile). Specifically, the average and high-end intakes of brazzein are 0.51 and 1.35 mg/kg bw/day, respectively, for non-diabetic adults; 0.56 and 1.79 mg/kg bw/day for diabetic adults; 0.85 and 1.98 mg/kg bw/day for non-diabetic children; and 1.34 and 1.82 mg/kg bw/day for diabetic children.

As discussed in more detail in Part 6, a no observed adverse effect level (NOAEL) of 978–985 mg/kg bw/day was established from a published 90-day oral toxicity study in rats (Lynch et al., 2023). A margin of safety (MOS) of approximately 493 can be estimated by comparing the NOAEL (*i.e.,* 978 mg/kg bw/day in male rats) with the highest intake (the most conservative) of 1.98 mg/kg bw/day of Farmtory™ Brazzein estimated above.[2] The test article in the 90-day study by Lynch et al. 2023 has a minimum brazzein content of 30%. Assuming brazzein is at the minimum content of 30%, the MOS can be conservatively calculated to be 148.[3] In addition, a NOAEL of 300 mg/kg bw/day of brazzein was established from an unpublished 90-day oral toxicity study in rats on Farmtory™ Brazzein sponsored by the notifier. This unpublished study is used as corroborative evidence to support the safety of brazzein. The MOS can be estimated to be 152.[4] Therefore, based on these estimates, the large MOS indicates that there is no safety concern on brazzein and supports the GRAS status of Farmtory™ Brazzein for its intended uses.

[2] 978 mg/kg bw/day ÷ 1.98 mg/kg bw/day = 493

[3] 978 mg/kg bw/day x 30% = 293 mg/kg bw/day
293 mg/kg bw/day ÷ 1.98 mg/kg bw/day = 148

[4] 300 mg/kg bw/day ÷ 1.98 mg/kg bw/day = 152

3.2 Dietary Intake Estimates of Minerals from Farmtory™ Brazzein

Farmtory™ Brazzein contains trace minerals and elements (Table 4). The dietary exposure of these minerals and elements has been estimated and compared with their respective Upper Limits established by the Institute of Medicine (IOM)/the National Academy of Medicine (NAM).

As the worst-case estimate, the dietary intake of minerals in Farmtory™ Brazzein was assessed based on the highest estimated absolute intake of 107.4 mg/day from diabetic adults (90th percentile).[5] The intakes of Farmtory™ Brazzein for children based on body weight are higher (i.e., 1.98 and 1.82 mg/kg bw/day). However, the absolute intakes are lower due to their lower body weight. We therefore use the highest absolute intake of 107.4 mg/day (most conservative) to calculate the mineral intakes from the intended uses of Farmtory™ Brazzein (Table 6). As shown in Table 6, the estimated mineral intakes under the proposed conditions of use remain well below their respective ULs established by IOM.

For sodium, Farmtory™ Brazzein contributes up to 0.43 mg/day to the diet, significantly lower than the FDA's recommended intake of 2,300 mg/day for individuals aged 14 and older. The contributions of other minerals are negligible, and their intakes from Farmtory™ Brazzein are not expected to pose any safety concerns.

Table 6 Mineral Intake from Farmtory™ Brazzein and Comparison to ULs by IOM

Element	Mineral in Farmtory™ Brazzein (ppm or mg/kg)	Mineral Intake from Farmtory™ Brazzein (mg/day)[a]	UL (mg/day)[b]
Phosphorus（P）	28.83	0.0031	4,000
Calcium（Ca）	32.2	0.0035	2,500
Copper（Cu）	3.9	0.00042	10,000
Iron (Fe)	17.8	0.0019	45
Potassium（K）	2.4	0.00026	NA
Magnesium（Mg）	6.1	0.00066	350
Manganese（Mn）	0.3	0.000032	11
Sodium（Na）	4011.4	0.43	NA
Zinc （Zn）	ND	ND	40

5 1.79 mg/kg bw/day x 60 kg bw = 107.4 mg/day

ND = not detected; UL = tolerable upper limit intake.

a Calculated based on the highest estimated absolute intake of 107.4 mg/day. For example, Phosphorus: 28.83 mg/kg ÷ 10^{-6} x 107.4 mg/day = 0.0031 mg/day.

b Value represents the tolerable upper intake level for male adults (age ≥18 years) as established by IOM/NAM. No tolerable upper intake level has been established by IOM/NAM for sodium and potassium due to insufficient evidence of a defined threshold.

PART 4 SELF-LIMITING LEVELS OF USE

Farmtory^{TM} Brazzein produced from fermentation has self-limiting levels related to the desired sweetness intended for a particular food or beverage product, just like other high intensity sweeteners. As such, the use of Farmtory^{TM} Brazzein as a general-purpose sweetener in foods is self-limiting based on its organoleptic properties.

PART 5 EXPERIENCE BASED ON COMMON USE IN FOOD BEFORE 1958

Not applicable.

PART 6 NARRATIVE AND SAFETY INFORMATION

6.1 Introduction

The subject of this GRAS Notice is Farmtory™ Brazzein, a protein-based sweetener manufactured through microbial fermentation using a bioengineered strain of *K. phaffii*. This process involves a safe strain lineage of *K. phaffii*, a non-pathogenic and non-toxigenic species genetically engineered to express the gene encoding brazzein. Brazzein, the active ingredient, was originally identified from the naturally occurring plant *Pentadiplandra brazzeana* and synthesized *de novo* before introduction into the production organism using standard biotechnology techniques.

The production process complies with current Good Manufacturing Practice (cGMP) and Hazard Analysis and Critical Control Points (HACCP), which includes fermentation, followed by concentration, purification, and drying into a powder consisting of ≥ 90% total protein (w/w) and ≥ 90% brazzein (w/w). All raw materials and processing aids used in the manufacturing process are safe and suitable standard ingredients, and are used in in accordance with appropriate U.S. regulations. The specifications and batch analyses of Farmtory™ Brazzein demonstrate and confirm the purity and safe nature of the GRAS substance, for example, free of heavy metals and microbiological impurities.

Therefore, the safety evaluation of Farmtory™ Brazzein focuses on two aspects: the safety of the production organism (modified *K. phaffii*) and the safety of the end product (brazzein).

6.2 Safety of the Production Strain *Komagataella phaffii*

MicroFarmtory conducted a comprehensive search of the publicly available scientific literature on the production strain *K. phaffii* through July 2025 and did not identify any safety concern.

6.2.1 Safety Assessment of Production Strain

K. phaffii has a long history of safe use in industrial scale protein production for food use. It is a well-recognized safe and suitable organism for production of food ingredients and is not capable of producing toxic metabolites when used for food protein production.

The safety of Farmtory™ Brazzein production strain was evaluated using principles commonly applied to the safety assessment of microbially derived enzyme preparations(FAO/WHO, 2020; Pariza & Johnson, 2001; Sewalt et al., 2016). This evaluation included assessments of the strain's pathogenicity, toxigenicity, antimicrobial resistance, and the genetic modifications involved in developing the strain. These elements are discussed below.

The production strain, *K. phaffii* GS115, was derived from the host strain *K. phaffii* NRRL Y-11430, and is classified as a biosafety level 1 (BSL1) organism by the ATCC. GS115 and NRRL Y-11430 are genomically similar and are well-documented in GRNs 737, 967, 1001, 1056, 1142, and 1167. *K. phaffii* (formerly known as *P. pastoris*) has an established history of safe use in food production, particularly for food enzymes and other ingredients (U.S. FDA, 2022d).In the European Union, *K. phaffii* (referred to as *Komagataella pastoris*) has qualified presumption of safety (QPS) status for food production based on its inability to produce toxic metabolites under standard food processing conditions (EFSA, 2018). *K. phaffii* is not classified as a pathogen by the European Commission, the National Institute of Allergy and Infectious Diseases, or the U.S. FDA (CE, 2000; NIAID, 2018; U.S. FDA, 2022b). In the United States, *P. pastoris* is permitted as a protein source in broiler feed at levels up to 10% (U.S. FDA, 2022c).

A comprehensive search of the literature did not yield any reports of pathogenicity or toxigenicity associated with *K. phaffii*. The evidence indicates that *K. phaffii* is a non-pathogenic, non-toxigenic species, a conclusion broadly accepted by the scientific community given its extensive use as a production organism for food ingredients. Furthermore, publicly available information on strains GS115 and NRRL Y-11430, as documented in previous GRAS notices (GRNs 737, 967, 1001, 1056, 1142, and 1167) for various food enzymes and ingredients that received "no questions" letters from the FDA, supports the safety and suitability of these strains for food production.

6.2.2 Genetic Modifications and Stability of the Production Strain

As described in Section 2.2.1, Farmtory^{TM} Brazzein production strain was bioengineered to express the gene encoding brazzein using standard biotechnology techniques. The strain has been confirmed to lack antibiotic resistance genes and residual expression plasmids. The synthetic gene encoding brazzein is the only secreted exogenous genetic material introduced into *K. phaffii* GS115. This gene is well-characterized, and bioinformatics analyses have confirmed it encodes the brazzein protein without introducing pathogenic or toxigenic properties into the host organism.

MicroFarmtory has also confirmed the genetic stability of the production strain throughout fermentation. Based on these findings, *K. phaffii* GS115 meets the criteria for a safe and suitable source organism, as outlined by Pariza and Johnson (Pariza & Johnson, 2001).

6.3 Safety of Brazzein

MicroFarmtory conducted a comprehensive search of the publicly available scientific literature on brazzein through July 2025 and did not identify any safety concerns. The studies include toxicological evaluations of brazzein-containing products and bioinformatics analyses to assess the allergenic and toxigenic potential of brazzein. The safety studies on brazzein are summarized below, all of which demonstrate the safety of Farmtory^{TM} Brazzein and support the GRAS conclusion of Farmtory^{TM} Brazzein for its intended uses.

A published study by (Lynch et al., 2023) on the safety evaluation of oubli fruit sweet protein (brazzein) derived from *Komagataella phaffii* serves as the pivotal study for the current GRAS notice. This study was also the pivotal study for GRN 1142 and GRN 1167 on Brazzein from fermentation that FDA reviewed and issued no questions letters. The study includes bacterial reverse mutation test (OECD, 2020), *in vitro* mammalian micronucleus test (OECD, 2016), and 90-day repeated-dose oral toxicity study in rats (OECD, 2018). All studies were conducted in accordance with Good Laboratory Practices (GLP) and appropriate OECD Test Guidelines. Results were also reported for a 14-day range finding study and an *in silico* allergenicity assessment of brazzein protein. The test article was described as a preparation of brazzein produced via fermentation with *K. phaffii* containing 85% total protein (≥ 30% brazzein w/w %), 5% carbohydrates, 10% ash, and ≤ 1% fat.

A recently published study investigated the toxicity of brazzein derived from *K. phaffii* for use as a sweetener in foods and beverages (Meetro et al., 2025). An *in vitro* digestibility study, an *in silico* allergenicity assessment, an *in vitro* reverse mutation assay, an *in vitro* mammalian micronucleus assay, and a 90-day oral toxicity study in rats following GLP and appropriate OECD test guidelines. The test material used in these toxicological tests was described as brazzein with a purity of 95.30%. Calculations of test article concentrations for all studies include a correction for purity using a factor of 1.049. Therefore, all doses in the toxicological tests are presented on the basis of the brazzein protein and not the whole ingredient.

In addition, another published study investigated the safety of a recombinant brazzein produced in *K. phaffii* (*P. pastoris*) in acute, subchronic, and chronic toxicity tests, allergenicity, and genotoxicity tests (Novik et al., 2023). The recombinant brazzein was stated to be identical to brazzein isolated from the ripe fruits of the *Pentadiplandra brazzeana* plant; the purity of the test material, however, was not reported. It was stated that the recombinant brazzein was purified using chromatography followed by 2 steps of ultrafiltration and lyophilization. The identification of protein was performed using expression construct sequencing followed by both gel electrophoresis and mass-spectrometry. The microbiological purity of sweet proteins was evaluated using bacteriological methods.

In addition to the published studies, MicroFarmtory sponsored a series of similar toxicity studies on Farmtory™ Brazzein, the subject of this GRAS notice. These studies included acute oral toxicity studies in mice and rats, 90-day oral toxicity study in rats, bacterial reverse mutation test, *in vitro* mammalian micronucleus test, and *in vitro* Chromosome aberration test of mammalian cells. These studies were conducted in accordance with GLP and National food safety standards of the People's Republic of China (or GB standards), which are essentially the same as the corresponding OECD Test Guidelines. Although these studies are unpublished, they corroborate the GRAS conclusion of Farmtory™ Brazzein based on the published study by Lynch et. al. (2023).

6.3.1 Published Safety Studies on Brazzein

6.3.1.1 Oral Toxicity Studies in Animals

Lynch et al. (2023) conducted a 14-day range finding study using four groups of adult CRL: Sprague Dawley rats (5/sex/group) which were fed diets prepared to contain 3000, 6000, and 12000 ppm of brazzein targeting daily intakes of 250, 500, and 1000 mg/kg bw/day, respectively (Lynch et al., 2023). The animals were observed at least once daily for viability, signs of gross toxicity, and behavioral changes, and weekly for a battery of detailed observations. Body weights were recorded twice during the acclimation period and on days 3, 7, 10, 14, and immediately prior to sacrifice. Individual feed consumption was also recorded to coincide with body weights. Feed efficiency and dietary intake were calculated. A gross necropsy was performed on all animals at study termination. No treatment related effects of brazzein were reported at up to nominal doses of 1000 mg/kg bw/day, including effects on mortality, clinical signs, body weight, body weight gain, or feed consumption. No treatment related abnormalities were noted upon macroscopic examination at necropsy. Based on the results of the 14-day study, a dose of 1000 mg/kg bw/day was selected as the high dose for the 90-day dietary toxicity study (below).

For the 90-day study, Sprague-Dawley rats [Crl:CD(SD)] (n=10/sex/group) were administered brazzein derived from K. phaffii via the diet at concentrations of 0 (control), 250 (low-dose), 500 (mid-dose), and 1,000 (high-dose) mg/kg bw/day for 90 days (Lynch et al., 2023). All study groups had ***ad libitum*** access to feed and water. The dietary concentrations achieved the targeted nominal intake values of 0, 250, 500, and 1,000 mg/kg bw/day, with actual values measured at 0, 245, 490, and 978 mg/kg bw/day for males and 0, 245, 493, and 985 mg/kg bw/day for females. Animal health was assessed three times daily, and weekly measurements of body weight and feed consumption were recorded. Ophthalmological examinations were performed pre-study and at Week 13. In the final week, urinalysis was conducted, and at study termination, hematology, blood chemistry, organ weight, macroscopic, and histopathological evaluations were carried out. Functional observational battery (FOB) tests were conducted once prior to test substance administration, during week 1, and monthly thereafter. Blood chemistry (including thyroid hormones), urinalysis, organ weights, and histopathology analyses were conducted per OECD 408 guidelines (OECD, 2018).

No mortalities occurred during the study, and there were no reports of clinical signs related to brazzein treatment. Hair loss (alopecia) was the main finding, but the incidence was spread throughout the control and treatment groups of both male and female rats, indicating this was not treatment related. Brazzein treatment also had no adverse effect on FOB measurements, body weight, body weight gain, feed consumption, or feed efficiency. There were no indications of an adverse effect of treatment on the results of the hematological, clinical chemistry, or urinalysis parameters measured. No biologically significant effects of brazzein treatment were reported on either absolute or relative organ weights, and histopathological examinations revealed no differences between treatment and control groups. Given the data reported, the authors concluded that the NOAEL for the study could be considered to be at least the maximum dose tested, namely, 978 mg/kg bw/day for males and 985 mg/kg bw/day for females.

Meetro et al. (2025) conducted a 14- day dose range–finding study with doses of 0 (purified water), 1000, 1500, or 2000 mg brazzein/kg bw/day administered via gavage to (Crl:WI[Han]) rats (5/sex/group) daily for 14 days (Meetro et al., 2025). No mortalities were observed at any timepoint of the study period, and no significant adverse effects attributable to brazzein were reported in any test parameter. As a result, a 90- day repeated-dose oral toxicity study was performed using doses of 500, 1000, and 2000 mg brazzein/kg body weight/day.

In the subsequent 90-day study, Wistar rats (10/sex/group) were administered 0 (purified water), 500 (low dose), 1000 (mid dose), or 2000 (high dose) mg/kg bw/day of brazzein via gavage once daily for 90 days. Five additional males and females were assigned to the control recovery group and high- dose recovery group. These animals received purified water or 2000 mg/kg body weight/day brazzein for the duration of the study period and were monitored during a subsequent 4-week recovery period with no treatment.

All animals survived until the end of the dosing period or recovery period. There were no brazzein-related clinical signs; all observations were transient and comparable in the treatment and control groups. No adverse ophthalmic findings were noted. Absolute body weights, body weight gain, and feed consumption were not significantly affected during the dosing period. Bioanalytical analyses did not reveal any significant brazzein absorption in rats.

No treatment-related effects on FOB tests were observed. A few findings were observed in the motor activity of high-dose males. Due to small magnitudes and transient nature of difference between the high-dose males and control group, in the absence of any correlating increased activity or behavioral changes for these animals in the corresponding FOB assessments, and in the absence of similar observations in high-dose females, these findings were considered by the authors to be incidental and attributed to biological variation.

At the end of the dosing period, no significant differences were observed for absolute and relative organ weights or for semen parameters in male rats. Following the recovery period, absolute and relative organ weights remained similar between treatment and control groups. No treatment-related, adverse macroscopic, or microscopic findings were identified in any treated animal from the dosing or recovery period.

Significant hematological changes observed in males (decreased red cell distribution width and lymphocyte count) remained within historical control data ranges, and similar changes were not observed in females. Therefore, the changes were not considered to be treatment-related. Significant findings from clinical chemistry analysis (i.e., glucose, phosphate, sodium, and AST) were all within the respective historical control data range and were only observed in one sex. The significant increase in serum bile acid was observed in mid- dose and high- dose females, and no similar changes were observed in males. No historical control data were available for comparison. However, there were no histopathological correlates for this finding in females. Based on these factors, findings from clinical chemistry analysis were not considered to be treatment-related.

A reduction in body weight gain was observed in high-dose animals during the recovery phase. The decrease in body weight change may be attributable to a slight increase in caloric content that the test animals received from brazzein during the treatment period, and the absence of brazzein in the recovery phase could have impacted the body weight gain. Nevertheless, the body weight gain change was observed during the recovery phase when the animals were no longer treated and thus was not considered an adverse effect of brazzein. There were no other statistically significant findings in any other study parameter.

As brazzein did not elicit any adverse effects in Wistar rats when administered for 90 days at doses up to 2000 mg/kg bw/day (the highest dose tested), the authors established 2000 mg/kg bw/day as the NOAEL.

Novik et al. (2023) investigated the safety of a recombinant brazzein produced in *K. phaffii* (*P. pastoris*) in acute, subchronic, and chronic toxicity tests, among other tests (Novik et al., 2023). The acute toxicity experiment on rats (males and females) and mice (males and females) established that the LD50 for brazzein was over 5000 mg/kg of body weight when administered intragastrically. Symptoms of animal intoxication and death were not observed for any tested dose during the entire experiment duration.

In the subchronic oral toxicity study, recombinant brazzein was administered via gavage to guinea pigs (9/sex/dose) at 0, 2.17, or 21.7 mg/kg bw/day for 21 days. No lethal cases or symptoms of intoxication were observed. Macroscopic examination and internal organ mass coefficient (organ-to-body weight ratio) revealed no pathological changes. During the experiment, no significant deviations of the biochemical parameters of experimental animals from the reference values or from the indicators of the control groups were found. The authors concluded that the absence of death or intoxication symptoms, the absence of negative effects on body weight gain, the internal organ, and the biochemical indices showed that daily intragastric administration of brazzein for 21 days did not demonstrate toxic effects on guinea pigs.

In the chronic oral toxicity study in rats, recombinant brazzein was administered via gavage to outbred rats (10/sex/dose) at 0 (vehicle control), 2.17, or 21.7 mg/kg bw/day for a period of 150 days. A positive control group was administered sucrose at 4824 mg/kg bw/day. The authors reported a statistically significant difference in relative body weight gain in male rats, but not in females, at 2.17 mg/kg bw/day (-13%) and 21.7 mg/kg bw/day (-16%) when compared to the control group. However, the dose groups administered sucrose (4824 mg/kg bw/day) also demonstrated a decrease in mean body weight gain (-12%) compared to the control group. As other indicators detected no toxic effects, the authors believe that the reduction in weight gain is the result of a slight decrease in feed intake because the administration of brazzein or sucrose caused the animals to reach satiety.

The administration of brazzein had no statistically significant effect on behavioral reactions. No pathological changes were observed in macroscopic and microscopic examinations of the liver, kidneys, lungs, heart, spleen, pancreas, lymph nodes (mesenteric), stomach, small and large intestines, brain, adrenal glands, thymus, testes (in males), and ovaries (in females) following administration of brazzein. Judging by the external signs, the reactions and behavior of animals, brazzein did not have an irritating effect on the gastrointestinal mucous, and the rats quickly calmed down after administration of the studied protein. This observation was confirmed by the results of histological examination of the gastrointestinal mucous, which indicated the absence of pathological changes after prolonged intragastric administration of brazzein.

In summary, these studies demonstrated that brazzein does not cause significant adverse effects in experimental animals. The NOAEL is at least the high dose tested in the 90-day studies that were conducted according to the OECD test guideline; for example, 978 mg/kg bw/day for males and 985 mg/kg bw/day for females (Lynch et al., 2023) and 2000 mg/kg bw/day (Meetro et al., 2025).

6.3.1.2 Genotoxicity Studies

Lynch et al. (2023) investigated the mutagenic potential of brazzein-53 using a bacterial reverse mutation assay with *Salmonella typhimurium* strains TA98, TA100, TA1535, and TA1537, as well as *Escherichia coli* WP2 uvrA, both with and without S9 metabolic activation (Lynch et al., 2023). The primary test conducted using the plate incubation method and a confirmatory test conducted using the preincubation method, both utilizing eight dose levels: 5,000, 1,580, 500, 158, 50, 15.8, 5, and 1.58 µg/plate) were conducted for all strains. Mean revertant colony counts for each treatment did not reach the level which would indicate that the treatment was mutagenic, which was defined for this study as a 2-fold or greater increase for strains TA98, TA

100, and WP2 uvrA and a 3-fold or greater increase for strains TA1535 and TA1537 when compared to the vehicle/negative control with or without S9 metabolic activation. As the test article did not induce signs of toxicity, the mean revertant colony counts for each strain tested with the vehicle/negative control were within the laboratory historical range, and the positive control substances caused expected substantial increases in revertant colony counts both with and without S9 metabolic activation, the test was considered valid, and the authors concluded that brazzein was not mutagenic under the conditions of the assay.

A micronucleus assay was undertaken to determine if brazzein induced structural or numerical chromosomal damage in primary human lymphocytes. An initial test for cytotoxicity indicated that the test article was not cytotoxic at the maximum suggested dose according to the OECD 487 guideline (OECD, 2016) of 2,000 µg/mL. Therefore, this was set as the maximum dose for the main experiments, consisting of 2,000, 1,000, 500, 250, 125, and 62.5 µg/mL including negative (cell culture media) and positive controls. Two separate experiments were undertaken, the first consisted of short term (4 hours) exposure of cells to the test article with and without S9 metabolic activation while the second consisted of long-term exposure (44 hours) without metabolic activation only. The highest 3 doses (2,000, 1,000, and 500 µg/mL) were chosen for microscopic analysis, and 1,000 cells were analyzed per dose per replicate. The test as a whole was considered to be valid as all conditions of validity were met: concurrent negative/vehicle controls were consistent with laboratory historical controls, concurrent positive controls induced a statistically significant increase in micronuclei, cell proliferation as measured by CBPI was ≥ 70% (≤ 30% cytostasis); and an adequate number of cells (≥ 500 cells per treatment) and concentrations (≥ 3) were analyzed. As the numbers of micronucleated cells exposed to brazzein were within or below the historical controls of the negative control and did not show a biologically relevant increase compared to the concurrent negative control and the $\chi 2$ test for trend did not show a concentration dependent statistically significant increase in micronuclei, the authors concluded that brazzein was not clastogenic or aneugenic under the conditions of the test.

Meetro et al. (2025) also conducted a bacterial reverse mutation test, using *Salmonella typhimurium* strains TA98, TA100, TA1535, and TA1537 and *Escherichia coli* strain WP2 uvrA pKM101 with or without S9 metabolic activation (Meetro et al., 2025). Strains were tested at various concentrations of up to 5000 µg/plate brazzein using the plate incorporation method (Experiment 1) and preincubation method (Experiment 2). In both experiments, brazzein did not produce any statistically significant increases in the number of revertant colonies under plate incorporation or preincubation methods in the bacterial reverse mutation test. No precipitation of the test material or cytotoxic effects were observed in either method. Therefore, not mutagenic under the conditions of the test.

In the *in vitro* micronucleus assay, concentrations for the main test were selected based on a preliminary range-finder experiment carried out using a maximum concentration of 2000 µg/mL to evaluate cytotoxicity. In the short-term assay without metabolic activation, no significant differences were observed in the frequency of MNBN cells between the treatment and vehicle control groups. The frequency of MNBN cells in the treatment groups fell within the historical vehicle control range data. In the short- term assay with metabolic activation, the frequency of MNBN cells at 500 and 2000 µg/mL were significantly higher than the concurrent vehicle controls. The increases demonstrated a significant linear trend; however, the results fell within the 95th percentile range of the historical control data and therefore were not considered to be biologically relevant. The statistically significant increases at 500 and 2000 µg/mL were attributed to a low concurrent vehicle control response. In the continuous assay, no significant differences were observed in the frequency of MNBN cells at 1000 and 2000 µg/mL compared to the vehicle control groups. At the lowest analyzed concentration (500 µg/mL), a statistically significant increase in MNBN cells was observed and was attributed to a low response in the concurrent vehicle control group. The frequency of MNBN cells in all treatment groups was within the 95th percentile range of the historical control data with no evidence of concentration-related increases. As such, the significant increase at 500 µg/mL was not considered to be biologically relevant. Confirmatory experiments for the short- term assay with metabolic activation and the continuous assay were performed using the same methodology as the main experiment to assess the re producibility of the statistically significant increases observed. The frequency of MNBN cells was not significantly different between the treatment and concurrent control groups in either assay. All results fell within the 95^{th} percentile range of the historical control data, and no concentration- related increases were observed.

Based on the above evidence, the authors concluded that brazzein was not genotoxic or mutagenic potential.

Similarly, Novik et al. (2023) conducted a bacterial reverse mutation assay using *S. typhimurium* strains TA98, TA97, and TA100, and demonstrated that recombinant brazzein ((50,000; 10,000; 2000; 400 and 80 µg/mL) exhibited no mutagenic activity (Novik et al., 2023). Additionally, recombinant brazzein showed no genotoxic effects in an *in vivo* micronucleus and chromosome aberration tests conducted on mice (CBAxD57B1) at doses up to 5000 mg/kg bw/day (Novik et al., 2023).

In conclusion, these studies demonstrated that brazzein is not genotoxic.

6.3.1.3 Allergenicity Studies

To assess the allergenicity potential of brazzein, Lynch et al. (2023) described searches conducted of the AllergenOnline database (www.allergenonline.org) maintained by the Food Allergy Research and Resource Program (FARRP) of the University of Nebraska-Lincoln and Allermatch (www.allermatch.org) database maintained by Wageningen University using the approaches described by Codex Alimentarius (Codex Alimentarius, 2009) and FAO/WHO (FAO/WHO, 2001). For both databases, sliding 80-amino acid alignment, exact 8 amino acid, and full-length amino acid searches were conducted using default settings (percent identity >35%) and the FASTA36 algorithm. Brazzein did not share any significant sequence homology matches with any known allergens from the AllergenOnline Database or Allermatch Database. All matches shared less than 35% identity in all 80-amino acid sliding window searches, and no 8-amino acid exact matches were identified with any known allergens from the AllergenOnline Database or Allermatch Database.

Similarly, Meetro et al. (2025) studied the digestibility of brazzein in an *in vitro* digestibility model with gastric and intestinal digestive phases (Meetro et al., 2025). The results showed that brazzein was not digested in the gastric phase but was partially digested in the intestinal phase, suggesting that the protein would not be absorbed into the systemic circulation as an intact protein. The partial digestion of brazzein further suggested that the protein would pose a low risk for allergenicity for consumers which is supported by the results of the *in silico* allergenicity assessment.

The results of the *in silico* allergenicity evaluation indicated that brazzein lacks allergenic potential. The recommended sequence homology searches (i.e., “sliding window” search of 80–amino and exact 8- amino acid match) using the latest version of the AllergenOnline database did not identify any positive matches with putative allergens. The full-length sequence homology search did not identify any matches with known allergens. The allergenic potential of brazzein-53 was evaluated in a similar *in silico* allergenicity assessment performed using the same allergen databases and homology searches (Lynch et al., 2023). As such, the authors concluded that brazzein-53 does not pose a risk for allergenicity. The article stated that brazzein-53 has been marketed as a sweetener in various foods and beverages in the United States since early 2024, and no case reports of allergenic reactions following consumption of food or beverage products containing this protein were identified in the scientific literature or reported to the FDA through the Center for Food Safety and Applied Nutrition Adverse Event Reporting System or FDA Adverse Event Reporting System, suggesting that brazzein is of low allergenic concern to consumers.

On the other hand, Novik et al. (2023) studied allergenic effects of brazzein on two species of laboratory animals: albino males guinea pigs and males outbred mice. Guinea pigs, which have a high sensitivity to allergens of different origin, were used to perform skin, conjunctival and nasal tests of brazzein for allergenic action, as well as for testing the indirect mast cell degranulation reaction. Mice were used to test the inflammatory response to concanavalin A (Novik et al., 2023). The results of skin, nasal, and conjunctival tests, the indirect reaction of mast cell degranulation in guinea pigs, and the inflammation reaction to concanavalin A in mice indicated that brazzein proteins was not allergenic.

6.3.2 Unpublished Safety Studies on Farmtory^{TM} Brazzein

In addition to the published studies summarized above, MicroFarmtory sponsored the following toxicological studies. These unpublished studies corroborate the GRAS status of brazzein produced via fermentation with *K. phaffii*. The test article for these studies is the subject of this GRAS notification, which meets all identity, specification, and production processes as described in Part 2.

6.3.2.1 Oral Toxicity Studies in Animals

Two acute oral toxicity tests were conducted, one in mice and the other in rats. The acute oral toxicity test of Farmtory^{TM} Brazzein was conducted in mice. The results showed that all animals remained active and healthy throughout the observation period, with no signs of poisoning, altered abnormal behavior, or deaths. The mice demonstrated normal weight gain, and no significant abnormalities were observed during the gross anatomical examination of the sacrificed animals. Since no gross observation was observed at necropsy, histopathological examinations were deemed unnecessary. Under the conditions of this test, the oral LD_{50} of Farmtory^{TM} Brazzein in both male and female mice was greater than 10,000 mg/kg body weight.

A second acute oral toxicity test was conducted to evaluate the safety of Farmtory™ Brazzein in SD rats. During the study, all rats remained active and exhibited no signs of poisoning or abnormal behavior. Their coats were healthy, and no deaths occurred. Weight gain was consistent across all groups, and no adverse effects were noted. Post-mortem examinations revealed no visible abnormalities in any organs, and as a result, no histological examinations were conducted. The oral LD_{50} of Farmtory™ Brazzein in male and female rats was greater than 10,000 mg/kg body weight.

Farmtory™ Brazzein was administered orally to Sprague-Dawley rat for a 90-day oral toxicity test at doses 0, 50, 100, and 300 mg/kg bw/day (calculated as pure brazzein). During the 90-day exposure period, the rats in each dose group were in good general condition, and no significant adverse signs were observed. There was no significant difference in body weight, feed intake, weight gain, and feed utilization rate at each time point between male and female rats in each dose group and the control group within 90 days. At the end of the test, no significant differences were found in hematological indices, blood biochemical indices, urine indices, wet weights of organs, and organ-to-body weight ratios between male and female rats in each dose group and the control group.

Histopathological examinations of rats in the high-dose group did not reveal any toxic histopathological changes in various organs caused by the test substance. Eye examinations of rats in both the control group and the high-dose group before and after exposure also showed no abnormalities. At the end of the 28-day follow-up observation, no related delayed toxic reactions were found in the animals of the follow-up observation group. There were no significant differences with biological significance in hematological indices, blood biochemical indices, urine indices, wet weights of organs, and organ-to-body weight ratios between the animals of the follow-up observation group and the control group.

Therefore, it was concluded that under the conditions of the 90-day oral toxicity study, the NOAEL was at least 300 mg/kg bw/day (calculated as pure brazzein), the high dose of the study.

6.3.2.2 Genotoxicity Studies

The Bacterial Reverse Mutation Test was conducted to evaluate the mutagenic potential of Farmtory™ Brazzein. Five test strains (TA97a, TA98, TA100, TA102, and TA1535) of *S. typhimurium* were used with and without S9 metabolic activation. The experiment utilized a plate inclusion method, where varying doses of Farmtory™ Brazzein, ranging from 2 µg/dish to 1250 µg/dish, were tested. The pre-test confirmed that Farmtory™ Brazzein formed a suspension in sterile distilled water and demonstrated no toxicity to the test strains. A dose of 1250 µg/dish was established as the maximum concentration for the formal test, as higher doses led to precipitation that interfered with colony counting. Two independent tests were conducted with incremental doses of Farmtory™ Brazzein, with three parallel plates prepared for each dose. The number of reverted colonies was counted and compared to the controls. The number of reverted colonies in the test plates was consistently less than twice that of the solvent control, with no observable dose-response relationship. Under the conditions of this study, Farmtory™ Brazzein demonstrated no mutagenic activity in *S. typhimurium*. This finding indicated that Farmtory™ Brazzein is not mutagenic.

An *in vivo* mammalian red blood cell micronucleus test was conducted to evaluate the genotoxicity of Farmtory^{TM} Brazzein. A total of 50 commonly used SPF-grade strain mice, equally divided between males and females, were housed in a controlled barrier environment. The test was designed with three dose groups (high: 3.33 g/kg body weight; medium: 1.67 g/kg body weight; low: 0.83 g/kg body weight). The test involved two oral gavage administrations, spaced 30 hours apart, with each group receiving 10 mL/kg body weight of their assigned test substance. Six hours after the second administration, the animals were euthanized, and bone marrow samples were collected from the femur. The bone marrow cells were prepared into smears, stained using Giemsa, and analyzed under a microscope to assess the frequency of micronucleus-containing polychromatic erythrocytes (PCE) and the proportion of PCE in total erythrocytes. Statistical analysis revealed no significant difference in the micronucleus frequency between the treated groups and the solvent control group ($P > 0.05$). The proportion of PCE in total erythrocytes was maintained at levels exceeding 20% of the control, indicating that bone marrow proliferation was unaffected. It was concluded that Farmtory^{TM} Brazzein did not induce an increase in the micronucleus frequency in mammalian polychromatic erythrocytes under the conditions of this study. These results indicated that Farmtory^{TM} Brazzein did not induce chromosomal damage in mammalian cells.

In vitro chromosome aberration test of mammalian cells was conducted to evaluate the potential genotoxicity of Farmtory^{TM} Brazzein. Cytotoxicity testing was conducted as a preliminary step to determine appropriate dose levels for the main test. Farmtory^{TM} Brazzein was tested at concentrations of 1250, 2500, and 5000 µg/mL in both the presence and absence of a S9 metabolic activation system. In the main chromosome aberration test, Farmtory^{TM} Brazzein was tested at the same concentrations (1250, 2500, and 5000 µg/mL) under conditions with and without S9 activation for 4 hours. Cells were treated, harvested, and stained for chromosomal analysis. A total of 100 metaphase cells per group were examined for chromosomal aberrations. Statistical analysis (χ^2 test) revealed no significant difference in aberration rates between the Farmtory^{TM} Brazzein -treated groups and the negative control group ($p > 0.05$). An extended test was conducted, which involved a 24-hour exposure of CHL cells to Farmtory^{TM} Brazzein without S9 mix at the same concentrations of the main test. No significant increase in chromosomal aberrations was observed compared to the negative control ($p > 0.05$). The results of both studies indicated that Farmtory^{TM} Brazzein did not induce chromosomal aberrations in mammalian cells under the tested conditions. Therefore, Farmtory^{TM} Brazzein was concluded to be non-genotoxic based on the chromosome aberration test.

6.3.2.3 Allergenicity Assessment

Farmtory^{TM} Brazzein, produced by fermentation, does not contain any protein from the 9 major food allergens as specified by the Federal Food, Drug, and Cosmetic Act. GRN 1142 and GRN 1167 on brazzein from fermentation by *K. phaffii* conducted a comprehensive analysis to assess the allergenic potential of brazzein and residual host cell proteins identified in the finished product. The results demonstrated no concern with regard to allergenicity related to the notified brazzein products. Similarly, MicroFarmtory utilized well researched and accepted allergenicity databases and determined that there was no allergenicity concern for Farmtory^{TM} Brazzein.

The Food and Agriculture Organization of the United Nations (FAO), the World Health Organization (WHO), and Codex Alimentarius recommend a stepwise approach for evaluating protein allergenicity. This method begins by assessing the source of the protein, followed by an evaluation of the protein itself for sequence homology with known allergenic proteins (Codex Alimentarius, 2009; FAO/WHO, 2001). A protein is considered to have significant sequence homology if an 80-amino acid sliding window (e.g., segments 1–80, 2–81, 3–82) derived from its full-length amino acid sequence shares more than 35% identity with a known allergen or if there is an exact match of six contiguous amino acids. In such cases, potential cross-reactivity between the proteins may exist (Codex Alimentarius, 2009; FAO/WHO, 2001). The 80-amino acid window corresponds to the typical domain size of a protein, which may contain epitopes responsible for antibody binding. If sequence homology is not identified, the protein is unlikely to be allergenic or cross-reactive with known allergens (Codex Alimentarius, 2009).

Although six to eight contiguous amino acids might represent the minimum length of linear immunoglobulin E (IgE) binding epitopes and T cell epitopes in peanut allergens (Ara h 1 and Ara h 2) (Abdelmoteleb et al., 2021; Herman et al., 2009; Ladics, 2019), their predictive value for potential allergenicity is limited, as such matches often yield false positives (Ladics, 2019). The allergenicity potential of brazzein was investigated using the stepwise approach outlined by FAO/WHO (FAO/WHO, 2001) and Codex Alimentarius (Codex Alimentarius, 2009). No residual host cell protein was investigated since no host cell protein met the cut-off identification criteria of 1% relative abundance. Farmtory™ Brazzein Sequence homology searches were conducted against the curated AllergenOnline (www.allergenonline.org) and Allermatch (https://allermatch.org) databases.

To evaluate whether brazzein in Farmtory™ Brazzein shares amino acid sequence homology with known allergens, suggesting potential allergenic cross-reactivity, a search of the AllergenOnline database (Version 22) maintained by the Food Allergy Research and Resource Program at the University of Nebraska-Lincoln was conducted. The 80-amino acid alignment search, using default settings (>35% identity) and the FASTA36 algorithm, identified no matches exceeding 35% identity. This suggests that brazzein in Farmtory™ Brazzein is unlikely to have allergenic cross-reactivity. Despite the possibility of false positives, a search for exact eight-amino acid matches was conducted. No such matches were found.

A full-length sequence homology search of brazzein in Farmtory™ Brazzein against the AllergenOnline database using default settings (search method: Full Fasta 36, E-value cutoff: 1, Max Alignments: 20) revealed one match to a putative allergen, the basic form of pathogenesis-related protein 1, from peach (*Prunus persica*) with >50% identity. However, the E-value (0.9) and bit-score (24.4) indicated low statistical significance. The literature suggests that allergenic cross-reactivity between proteins with less than 50% similarity is rare and typically requires >70% identity (Aalberse, 2000). Matches with E-values below 0.001 and bit-scores above 40 are generally considered significant (Pearson, 2013). Abdelmoteleb et al. (Abdelmoteleb et al., 2021) further indicated that E-values below 10^-7 suggest biologically relevant similarity. Since the identified match exhibited less than 70% identity with an E-value of 0.9 and a bit-score of 24.4, it was deemed insignificant and unlikely to reflect allergenic cross-reactivity of the brazzein in Farmtory™ Brazzein.

Additional *in silico* allergenicity analyses were conducted using Allermatch, a database maintained by Wageningen University, which incorporates data from UniProtKB (www.uniprot.org/docs/allergen), the WHO and the International Union of Immunological

Societies allergen nomenclature list (www.allergen.org/), and the Comprehensive Protein Allergen Resource (https://comparedatabase.org). The 80-amino acid alignment search using default settings (>35% identity) and the FASTA algorithm similarly found no matches between brazzein in Farmtory^{TM} Brazzein and putative allergens. An eight-amino acid exact match search also yielded no results.

A full-length sequence search of brazzein against the Allermatch database identified >50% identity with 16 sequences. However, the statistical significance of these alignments was low, as all E-values exceeded 0.001 (lowest E-value: 1.7), and all bit-scores were below 40 (highest bit-score: 23.6). These alignments were therefore not considered significant and unlikely to indicate allergenic cross-reactivity.

6.3.2.4 Toxigenicity

To assess whether brazzein within Farmotry Brazzein shares significant sequence homology with known protein toxins, sequence homology searches were performed using the Basic Local Alignment Search Tool (BLAST). The amino acid sequence of brazzein in Farmtory^{TM} Brazzein was compared against protein sequences from a curated database of 7,970 (Animal toxin annotation project | UniProt help | UniProt, updated on April 18, 2023) animal venom proteins and toxins maintained by UniProt.

The BLAST searches utilized an algorithm with a word size of 6, an expect threshold of 0.001, and the BLOSUM62 scoring matrix with default gap costs and composition adjustments. Structural homology or similarity was evaluated based on criteria established by Pearson (Pearson, 2013), which indicate that homologous sequences sharing more than 40% identity are highly likely to have functional similarity. Additionally, an E-value of <0.001 is considered a reliable indicator of sequence homology. Bit-scores were also used as an alternative metric, as they provide a more robust indicator of significant sequence homology. A bit-score of 50 is "almost always significant," while a bit-score of 40 is only significant (E-value < 0.001) for protein databases with fewer than 7,000 entries (Pearson, 2013). Based on these criteria, no significant similarity to venom proteins or toxins was detected (E-values < 0.001) in the homology searches.

In summary, the results from published studies by Lynch et al., 2023, Meetro et al., 2025, and Novik et al., 2023 demonstrated that brazzein is not mutagenic/genotoxic, not allergenic, and not toxic. The NOAEL for the 90-day study is the maximum dose tested, namely, 978 mg/kg bw/day for males and 985 mg/kg bw/day for females (Lynch et al., 2023), and 2000 mg/kg bw/day (Meetro et al., 2025). The results from unpublished studies sponsored by MicroFarmtory on Farmtory^{TM} Brazzein are in line with those from the published studies. Farmtory^{TM} Brazzein is not mutagenic/genotoxic, not allergenic, and not toxic. The NOAEL for the unpublished 90-day study on Farmtory^{TM} Brazzein is 300 mg/kg bw/day (calculated as pure brazzein), the high dose tested.

6.4 Safety Discussion

Brazzein is isolated from the ripe fruits of the wild West African plant *Pentadiplandra brazzeana*. Indigenous people have used it for centuries as a sweetener for drinks and food

(Hellekant & Danilova, 2005). MicroFarmtory produces Farmtory™ Brazzein by fermentation using *K. phaffii* (formerly known as *P. pastoris*), which has an established history of safe use in food production. *K. phaffii* is a well-recognized safe and suitable organism for production of food ingredients and is not capable of producing toxic metabolites when used for food protein production. *K. phaffii* is classified as BSL1 by the ATCC and has the QPS status in the EU (EFSA, 2018). The production strain, *K. phaffii* GS115, was derived from the host strain *K. phaffii* NRRL Y-11430. GS115 and NRRL Y-11430 are genomically similar and are well-documented in GRNs 737, 967, 1001, 1056, 1142, and 1167 that FDA reviewed and issued no questions letters. The host strain does not contain any native plasmids or antibiotic resistance genes.

Farmtory™ Brazzein is manufactured under cGMP for food and meets Hazard Analysis and Critical Control Points (HACCP) requirements. All raw materials and processing aids used in the manufacturing process are safe and suitable standard ingredients, that are food grade and free of allergens. Farmtory™ Brazzein does not contain any protein from the 9 major food allergens as specified by the Federal Food, Drug and Cosmetic Act.

The results from published studies (Lynch et al., 2023; Meetro et al., 2025; Novik et al., 2023) demonstrated that brazzein was not mutagenic/genotoxic, not allergenic, and not toxic. The NOAEL from one 90-day study was the maximum dose tested, namely, 978 mg/kg bw/day for males and 985 mg/kg bw/day for females (Lynch et al., 2023), and was 2000 mg/kg bw/day in a second study (Meetro et al., 2025). The results from unpublished studies sponsored by MicroFarmtory on Farmtory™ Brazzein are in line with those from the published studies. In other words, Farmtory™ Brazzein was not mutagenic/genotoxic, not allergenic, and not toxic. The NOAEL for the unpublished 90-day study on Farmtory™ Brazzein was 300 mg/kg bw/day (calculated as pure brazzein), the high dose tested.

Farmtory™ Brazzein is intended for use as a general-purpose sweetener in foods and beverages, excluding infant formula and meat and poultry products. Part 3 Dietary Exposure applied the Renwick method, a well-established model, to estimate dietary exposure to Farmtory™ Brazzein. Brazzein is 500–2000 times sweeter than sucrose, depending on the concentration of the comparison solution (Ming & Hellekant, 1994). Assuming Farmtory™ Brazzein is 500 times as sweet as sucrose, the highest estimated intake of Farmtory™ Brazzein is 1.98 mg/kg bw/day for non-diabetic children at the high-end (or ≥ 90th percentile).

As summarized above, a NOAEL of 978–985 mg/kg bw/day was established from a published 90-day oral toxicity study in rats (Lynch et al., 2023). This study was the pivotal study for GRN 1142 and 1167 that FDA reviewed and issued no questions letter. It also serves as the pivotal study for the current GRAS notice. A MOS can be calculated to be approximately 493 (978 mg/kg bw/day ÷ 1.98 mg/kg bw/day = 493). The test article in the 90-day study (Lynch et al., 2023) has a minimum brazzein content of 30%. Assuming brazzein is at the minimum content of 30%, the MOS can be conservatively calculated to be 148 (978 mg/kg bw/day x 30% = 293 mg/kg bw/day; 293 mg/kg bw/day ÷ 1.98 mg/kg bw/day = 148).

A NOAEL of 2000 mg/kg bw/day was derived from a recently published study (Meetro et al., 2025). Given that this publication became available online in May 2025 and there is no record of FDA reviewing it, we consider it one of the corroborative studies. Nevertheless, using this

NOAEL to compare with the EDI of 1.98 mg/kg bw/day would yield an MOS of 1010 (2000 mg/kg bw/day ÷ 1.98 mg/kg bw/day = 1010).

Further, a NOAEL of 300 mg/kg bw/day of brazzein was established from an unpublished 90-day oral toxicity study in rats on Farmtory^{TM} Brazzein sponsored by the notifier. The MOS can be estimated to be 152 (300 mg/kg bw/day ÷ 1.98 mg/kg bw/day = 152).

These large MOSs indicate that brazzein poses no safety concern and support FarmtoryTM Brazzein's GRAS status for its intended uses.

Therefore, based on the safety of production strain *K. phaffii*, the manufacturing process and materials used, the safety studies and allergenicity assessment conducted on brazzein products and Farmtory^{TM} Brazzein, and the large MOS calculated, MicroFarmtory concludes that Farmtory^{TM} Brazzein is GRAS for its intended uses in conventional food and beverage products.

6.5 GRAS Panel Evaluation

MicroFarmtory has concluded that Farmtory^{TM} Brazzein is GRAS for use in conventional food and beverage products, as described in Section 1.3, on the basis of scientific procedures. This GRAS conclusion is based on data generally available in the public domain pertaining to the safety of brazzein, as discussed herein, and on consensus among a panel of experts (the GRAS Panel) who are qualified by scientific training and experience to evaluate the safety of food ingredients. The GRAS Panel consisted of the following qualified scientific experts: Professor Joseph Baumert (University of Nebraska Lincoln); Professor A. Wallace Hayes (Ph.D., DABT, FATS, ERT, University of South Florida College of Public Health), and Professor Emeritus Michael W. Pariza (University of Wisconsin Madison). The GRAS Panel members confirmed they have signed a statement of no conflict of interest.

The GRAS Panel, convened by MicroFarmtory, independently and critically evaluated all data and information presented herein and also concluded that Farmtory^{TM} Brazzein is GRAS for use in conventional food and beverage products as described in Section 1.3, based on scientific procedures. A summary of data and information reviewed by the GRAS Panel and evaluation of such data as it pertains to the proposed GRAS uses of Farmtory^{TM} Brazzein are presented in Appendix A.

6.6 Conclusion

MicroFarmtory manufactures Farmtory^{TM} Brazzein, a high-intensity sweetener, for use in food and beverage products, as outlined in this notification. MicroFarmtory has determined that Farmtory^{TM} Brazzein produced from fermentation by *K. phaffii* is GRAS for the intended use in food based on the following:

- Farmtory^{TM} Brazzein is manufactured under cGMP for food and meets Hazard Analysis and Critical Control Points (HACCP) requirements. All raw materials and processing aids used meet appropriate food grade specifications;

- Farmtory™ Brazzein contains a high level of brazzein protein (≥ 90%). Heavy metals and microbiological limits are either absent (not detected) or below toxicological and regulatory limits;
- Safety of the production organism, *K. phaffii* (formerly *P. pastoris*), is well established. It has a long history of safe use in the industrial scale production of proteins used broadly in a number of foods. Data support that the organism is non-pathogenic and non-toxigenic;
- Safety of brazzein is demonstrated by a series of published toxicological studies and allergenicity assessment on brazzein from fermentation, corroborated with unpublished studies on Farmtory™ Brazzein. These studies demonstrated that brazzein is not mutagenic/genotoxic, not allergenic, and not toxic, and the NOAELs are the highest doses in the studies.
- The intended use of Farmtory™ Brazzein and the estimated intake based on the Renwick method, a well-established model for estimating intake of high-intensity sweeteners, yield large MOS when compared with the NOAEL from the 90-day study, indicating that there is no safety concern and supporting the GRAS status of Farmtory™ Brazzein for its intended uses.

Based on the data presented, MicroFarmtory concludes that Farmtory™ Brazzein is GRAS for use in conventional food and beverage products under the proposed conditions of use. Therefore, Farmtory™ Brazzein may be marketed and sold in the U.S. without the promulgation of a food additive regulation under Title 21, Section 170.3, of the Code of Federal Regulations.

PART 7 LIST OF SUPPORTING DATA AND INFORMATION

7.1 References

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7.2 List of Appendices

A. GRAS Panel Evaluation Statement
B. Protein Identification Report
C. Farmtory^{TM} Brazzein Specification
D. Farmtory^{TM} Brazzein COA report batch 20240730
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GRAS Panel Evaluation of Farmtory^{TM} Brazzein for Use in Conventional Food and Beverage Products

July 15th, 2025

INTRODUCTION

MicroFarmtory convened a panel of independent scientists (the GRAS Panel), each possessing extensive expertise in food safety evaluation, toxicology, microbiology, and biochemistry, to critically assess the safety of Farmtory^{TM} Brazzein. The GRAS Panel was assembled in accordance with the U.S. Food and Drug Administration (FDA)'s guidance on best practices for convening a GRAS Panel.

The GRAS Panel comprised the following scientific experts:

- **Professor Joseph Baumert**, University of Nebraska-Lincoln
- **Professor A. Wallace Hayes, Ph.D., DABT, FATS, ERT**, University of South Florida College of Public Health
- **Professor Emeritus Michael W. Pariza**, University of Wisconsin-Madison

Each panelist conducted an independent, comprehensive review of the available scientific data regarding the identity, purity, manufacturing process, dietary exposure, and safety of Farmtory^{TM} Brazzein. Their assessment included critically evaluating relevant toxicological, allergenicity, and genotoxicity studies and comparisons with previously FDA-reviewed sweet proteins.

SUMMARY AND BASIS FOR GRAS

Farmtory^{TM} Brazzein is a sweet-tasting protein naturally found in the West African Oubli fruit (*Pentadiplandra brazzeana*). MicroFarmtory produces this protein via precision fermentation using a genetically modified strain of *Komagataella phaffii* (formerly *Pichia pastoris*), which has been optimized for enhanced protein expression. The final product, Farmtory^{TM} Brazzein, consists of $\geq$ 90% brazzein protein.

Manufacturing Process and Quality Control

The GRAS Panel carefully reviewed the production process of Farmtory^{TM} Brazzein, which adheres to cGMP and HACCP principles. The process includes:

- Fermentation using a genetically modified *K. phaffii* strain
- Solid-liquid separation to isolate the target protein
- Multi-step purification involving filtration, chromatography, and diafiltration
- Drying into a powder format

Product specifications include limits for proximate composition, heavy metals, and microbiological contaminants, all of which comply with recognized food safety standards. Batch analyses confirm a consistent, high-purity product.

Safety Assessment

Production Organism Safety

- The production strain *K. phaffii* GS115 is derived from *K. phaffii* NRRL Y-11430, a Biosafety Level 1 organism with a well-documented history of safe use.
- No antibiotic resistance genes or exogenous pathogenic factors were introduced.
- Genetic modifications were confirmed stable before and after fermentation.

Toxicological and Genotoxicity Studies

The GRAS Panel reviewed both published and unpublished studies, for example:

1. Bacterial Reverse Mutation Test: No mutagenic effects were observed.
2. In Vitro Micronucleus Test: No evidence of chromosomal damage.
3. 90-Day Oral Toxicity Study (Sprague-Dawley rats): No treatment-related toxicological effects up to 1000 mg/kg body weight/day.
4. Acute Oral Toxicity (Mice and Rats): LD_{50} > 10,000 mg/kg body weight, indicating non-toxic classification.

The NOAEL (No-Observed-Adverse-Effect Level) was determined to be 978 mg/kg body weight/day (males) and 985 mg/kg body weight/day (females) from the published 90-day study, yielding an estimated margin of safety > 100 under intended use conditions.

Allergenicity and Toxigenicity Assessment

- **Allergenicity:** Following the framework of the FAO/WHO and Codex Alimentarius, sequence database searches using both AllergenOnline and Allermatch indicated no significant sequence homology to known allergens.
- **Toxigenicity:** No structural similarities to known toxins (BLAST search against curated toxin databases).

Dietary Exposure and Intended Use

Farmtory™ Brazzein is intended for use as a high-intensity sweetener in conventional foods and beverages, excluding infant formula and meat and poultry products. The estimated daily intake (EDI) assessment, based on the Renwick method, indicated that projected exposures are below safety thresholds.

CONCLUSIONS OF THE GRAS PANEL

After an exhaustive review of the available data, including toxicology, allergenicity, dietary exposure, and product specifications, the GRAS Panel unanimously concludes that Farmtory™ Brazzein is Generally Recognized as Safe (GRAS) for its intended use based on scientific procedures.

It is the expert opinion of the GRAS Panel that other qualified scientists would concur with this conclusion. Therefore, Farmtory™ Brazzein may be marketed and used in conventional food and beverage products in the U.S. without the need for food additive regulation under 21 CFR §170.3.

Signature:

[redacted]

Joe Baumert, Ph.D.
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Director Food Allergy Research & Resource Program,
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[redacted]

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Michigan State University Center for Integrative Toxicology

[redacted]

Michael W. Pariza, Ph. D.
Emeritus Professor, Food Science
Emeritus Director, Food Research Institute
University of Wisconsin- Madison

Protein Identification Report

Project Type Protein Identification

Introduction

Proteomic is an emerging large scale and high throughout approach which can study comprehensive protein profile of biological sample. With the revolution of Mass spectrometry (MS) technology, Proteomic methodology has evolved rapidly in the past decade, and MS based proteomics has become an essential technology in the field of protein era. Wide Proteomic analysis of protein identification, quantification or post-translational modification (PTM) is playing a vital role in exploring underlying functional mechanism. In 2003, after the completion of the Human Genome Project (HGP), the Human Proteome Project (HPP) was launched in the same year, reflecting the scientific research field's emphasis on proteomics. What`s more, Human proteome blueprint was assembled over 10 years by multiple institutions and published in the journal Nature in 2014.

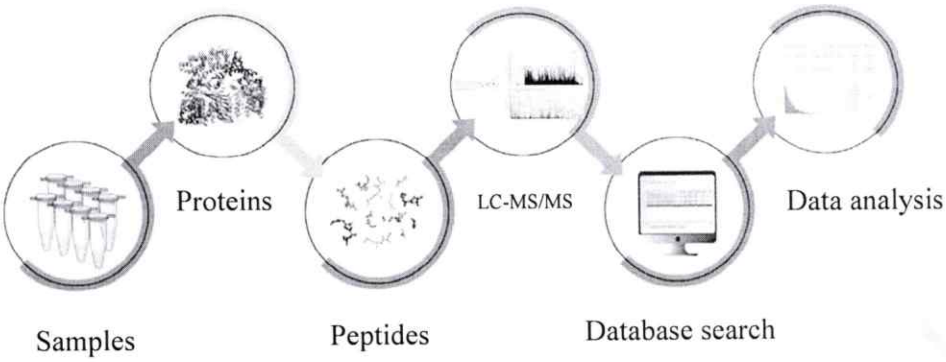
Currently, bottom to up is the most popular proteome research strategy, which involves sample preparation, LC-MS/MS analysis, and data analysis. The qualitative and quantitative information of the proteins in the samples can be obtained by analyzing the mass spectra data using related software.

Project information

1 Sample information

Number of samples	1
Sample name	Brazzein

2 Technical approaches



3 Result

1 Identification information

1.1 distribution of peptide length

The mass spectrometry (MS1) scan range is typically from 350 to 1500 m/z. After ionization, most peptide ions are charged with +2 (with +3, +4, etc., decreasing in abundance). The average molecular weight of an amino acid residue in a protein is about 110 Da, so most peptides detected by mass spectrometry are within the range of 7 to 27 amino acids.

The distribution of detected peptide lengths in this project is shown in the figure below. Peptides with a length of 7 to 27 amino acids account for 96.04%.

Figure1.1 distribution of peptide length

The horizontal axis represents the length of the identified peptides in terms of amino acid count, while the vertical axis indicates the number of peptides identified for each corresponding length.

1.2 distribution of peptide count

In proteomics, the "bottom-up" strategy is frequently used. The identification and quantification information about proteins is obtained directly from the peptide information detected by mass spectrometry. Each protein theoretically contains multiple digestible peptides, but the number of peptides that support the identification of each protein varies due to various factors such as peptide length, relative abundance, and ionization efficiency. Typically, the protein abundance is positive correlated with peptides count. Proteins identified by two or more peptides are generally considered to have a higher probability. The peptide count distribution graph represents the distribution of proteins with different numbers of peptides; high proportion of proteins with abundant peptides suggests reliable proteomics results at the protein level.

The distribution of the number of peptides detected in this project is shown in the figure below. The percentage of proteins with two or more peptides account for 58.33%.

1.3 distribution of Missed Cleavage Sites

Trypsin, an enzyme specifically hydrolyzes the peptide bonds at the C-terminus of arginine and lysine residues, converting the intact proteins into peptides of different lengths suitable for MS detection. However, protein tryptic-digestion efficiency is interfered with consecutive distribution of arginine or lysine in the protein sequence or PTM on the arginine or lysine side chains. Typically, most peptides will be detected with no missed cleavage sites, while little with one or two. A significantly high proportion of peptides with missed cleavage sites may indicate a lower efficiency of enzymatic digestion.

As shown in the figure below, the percentage of peptides without any missed cleavage sites accounts for 61.39%.

Figure 1.3 distribution of Missed Cleavage Sites

The horizontal axis represents the number of missed cleavage sites in the identified peptides. The vertical axis indicates the number of peptides containing the corresponding number of missed cleavage sites.

1.4 result statistics

In this project, 1 samples were carried on MS detection, and a total of 31 protein were identified. Partial result was shown in the below table, and detailed information can be acquired in the "summary.identification.xlsx" file.

Sample	Accession	Gene	Mw (kDa)	Peptides	Coverage	iBAQ (%)
Brazzein	target		6.370	12	96.2%	99.5416%
	A0A1B2JAS9	RPL40B	14.568	9	52.3%	0.2119%
	A0A1B2JDU3	ATY40_BA7503273	36.070	6	13.4%	0.0322%

The overview of protein, peptide, and spectrum identification for each sample is shown in the figure below.

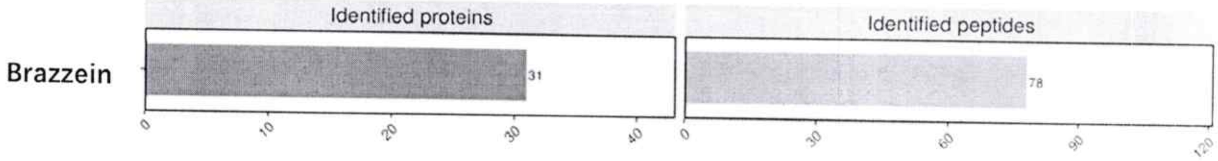


Figure 1.4-1 Sample Summary Chart

The one graphs exhibit the number of proteins (left) and peptides (right) identified in this project. The horizontal axis represents the count of statistics, and the vertical axis represents the sample name.

4 Materials and methods

4.1 sample preparation

The sample was mashed with a glass rod, followed by ddH2O, decolorizing solution (Coomassie staining: 50% ACN/100 mM NH4HCO3, pH=8.0; Silver staining: 30 mM K3Fe(CN)6/100 mM Na2S2O3) and ACN for decolorization. The sample was vortexed for 5 min, centrifuged and the supernatant was discarded. Protein reduction and alkylation were conducted with TCEP and CAA at 60°C for 30 min. The sample was vortexed for 5 min with ACN addition, centrifuged to discard the supernatant and then vacuum dried. An appropriate amount of trypsin was added according to the sample volume for overnight digestion at 37°C. The next day, the peptide extraction solution (60% ACN/5% formic acid) was added and then the mixture was sonicated for 10 min, centrifuged to remove the supernatant and vacuum dried. The C18 desalting column was used for desaltlination treatment. The sample was vacuum dried and stored at -20°C for later use.

4.2 MS analysis

All samples were analyzed on an UltiMate 3000 RSLCnano system coupled on-line with Q Exactive HF mass spectrometer through a Nanospray Flex ion source (Thermo). Peptide samples were injected into a C18 Trap column (75 μm*2 cm, 3 μm particle size, 100 Å pore size, Thermo), and separated in a reversed-phase C18 analytical column packed in-house with ReproSil-Pur C18-AQ resin (75 μm*25 cm, 1.9 μm particle size, 100 Å pore size). Mobile phase A (0.1% formic acid/3% DMSO/97% H2O) and mobile phase B (0.1% formic acid/3% DMSO/97% ACN) were used to establish the seperation gradient at a flow rate of 300 nL/min. The MS was operated in DDA top20 mode with a full scan range of 350-1500 m/z. AGC Target value for the full MS scan was 3E6 charges with a maximum injection time of 30 ms and a resolution of 60,000 at m/z 200. Precursor ion selection window was kept at 1.4 m/z and fragmentation was achieved by higher-energy collisional dissociation (HCD) with a normalized collision energy of 28. Fragment ion scans were recorded at a resolution of 15,000, an AGC of 1E5 and a maximum fill time of 50 ms. Dynamic exclusion was enabled and set to 30 s.

4.3 database search

MS raw data were analyzed with MaxQuant (V1.6.6.0) using the Andromeda database search algorithm. Spectra files were searched against the Pichia_pastoris protein sequence database(2020-07-25, 5034 entries) from Uniprot and Add.24041108.20240422.fasta from custom sequence files using the following parameters: variable modifications were set as Oxidation (M)-15.994915, Acetyl (Protein N-term)-42.010565, Deamidation (NQ)-0.984016, fixed modification was set as Carbamidomethyl (C); Enzyme digestion specificity was set to Trypsin/P with maximum 2 missed cleavages; Peptide mass tolerance in first search and main search were set to 20 ppm and 4.5 ppm; Fragment match tolerance was set to 20 ppm. Proteins could not be distinguished on the basis of unique peptides were merged by MaxQuant into one protein group. Search results were filtered with 1% FDR at both peptide and protein levels.

4.4 data analysis

Proteins denoted as reverse, or only identified by site were removed. The "proteinGroups.txt" output table was used for analysis on the protein level. Further bioinformatics analysis was conducted in R statistical programming environment.

5 Reagents and Suppllies

Number	Type	Chinese name	English name	Manufacturers	CAS
1	reagent	三(2-羧乙基)膦盐酸盐	Tris(2-carboxyethyl)phosphine hydrochloride (TCEP)	Sigma-Aldrich	C4706-10G
2	reagent	氯乙酰胺	2-Chloroacetamide (CAA)	Sigma-Aldrich	C0267-100G
3	reagent	胰蛋白酶	Trypsin	SignalChem	T575-31N-10
4	reagent	甲酸	Formic acid	Thermo Fisher Scientific	A118P-500ml
5	reagent	乙腈（HPLC）	Acetonitrile (ACN)	Thermo Fisher Scientific	A998-4
6	reagent	碳酸氢铵	Ammonium bicarbonate (NH4HCO3)	Fluka	09830-500G
7	reagent	铁氰化钾	Potassium ferricyanide (K3Fe(CN)6)	Sigma-Aldrich	702587
8	reagent	硫代硫酸钠	Sodium hyposulfite (Na2S2O3)	Sigma-Aldrich	217263
9	instrument	真空离心浓缩仪	SpeedVac	吉艾姆	CV200
10	instrument	漩涡混合器	Vortex Mixer	其林贝尔	VORTEX-5
11	instrument	掌上离心机	Mini Centrifuge	大龙	D1008E
12	instrument	干式恒温器	Block Heater	杭州米欧	DKT-100
13	instrument	微量台式冷冻离心机	Freezing centrifuge	Thermo Scientific	Micro17R

14	instrument	恒温混匀仪	Thermo mixer	杭州米欧	MTC-100
15	instrument	超声波清洗仪	Ultrasonic cleaner	洁盟	JP-040
16	instrument	UltiMate 3000 纳升高效液相色谱	UltiMate 3000 RSLCnano system	Thermo Scientific	UltiMate 3000 RSLCnano
17	instrument	Q Exactive HF 质谱仪	Q Exactive HF mass spectrometer	Thermo Scientific	Q Exactive HF
18	Supply	纳升 C18 捕集柱	Acclaim PepMap (75μm*2cm, C18, 3μm, 100Å)	Thermo	164535
19	Supply	纳升色谱柱 C18 填料	ReproSil-Pur 120 C18-AQ, 1.9μm)	Dr. Maisch	r119.aq.0001
20	reagent	水（LC-MS）	Water (LC-MS)	Thermo Fisher Scientific	W6-4
21	reagent	乙腈（LC/MS）	Acetonitrile (LC-MS)	Thermo Fisher Scientific	A955-4
22	reagent	二甲基亚砜	Dimethyl sulfoxide (DMSO)	Sigma-Aldrich	34869-100ML
23	reagent	甲酸（LC-MS）	Formic acid (LC-MS)	Fluka	60-006-17
24	software	MaxQuant (V1.6.6.0)	MaxQuant (V1.6.6.0)	Max-Planck Institute for Biochemistry	www.maxquant.org
25	database	UniProt	UniProt		www.uniprot.org
26	software	R	R		www.r-project.org

6 References

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Introduction of high-resolution MS Q Exactive HF

2. MaxQuant Software for Ion Mobility Enhanced Shotgun Proteomics. Mol Cell Proteomics, 2020. 19(6): p. 1058-1069.

MaxQuant is the most used free software for proteomics data analysis, and it has been updated to support data from the timsTOF Pro data.

3. UniProt: the universal protein knowledgebase in 2021. Nucleic Acids Res, 2021. 49(D1):p. D480–D489.

UniProt is a comprehensive and integrated database of protein sequences, functions, and analyses.

Farmtory Brazzein Specification

Parameter	Method	Specification
Appearance	GB/T 10220-2012	Yellowish brown powder
Odor	GB/T 10220-2012	Characteristic
Taste	GB/T 10220-2012	Intense Sweetness
Total protein (w/w%)	AOAC 930.29	≥90%
Brazzein (w/w%)	HPLC (QC-00-01)	≥90%
Loss on Drying (w/w%)	AOAC 930.04	≤10%
Fat (w/w%)	AOAC 996.06	≤1%
Ash (w/w%)	AOAC 930.30	≤5%
Carbohydrates (w/w%)	FDA 21 CFR 101.9	≤5%
Arsenic	AOAC 2015.01 ICP-MS	≤0.2 ppm
Lead	AOAC 2015.01 ICP-MS	≤0.1 ppm
Cadmium	AOAC 2015.01 ICP-MS	≤0.1 ppm
Mercury	AOAC 2015.01 ICP-MS	≤0.1 ppm
Total plate count	AOAC 990.12	≤5,000 CFU/g
Total Coliforms	AOAC 991.14	≤10 CFU/g
Yeast & Molds	AOAC 997.02	≤100 CFU/g
E.Coli	AOAC 991.14	≤10 CFU/g
Salmonella	AOAC 967.26	Not detected in 25g
Listeria	AOAC 993.12	Not detected in 25g

QC: [redacted]
Date: 2024.04.01

QA: [redacted]
Date: 2024.04.01

File Code: QC-01
Issue Date: Mar. 01,2024

Version Number: 01

Address: Suzhou Aquafarmtory Biotechnology LTD 68 Xinqing Road, Suzhou, Jiangsu,PRC,China
Contact: webpr@aquafarmtory.com
Website: www.aquafarmtory.com

水滴农厂 Farmtory

CERTIFICATION OF ANALYSIS			
Product Name	Farmtory Brazzein	Batch No.	20240730
Date of Manufacture	Jul. 30,2024	Quantity	1000g
Date of Certificate	Aug. 7,2024	Best Before Date	Jan. 30,2026
Test	**Specification**	**Method**	**Result**
Appearance	Yellowish brown powder	GB/T 10220-2012	Yellowish brown powder
Odor	Characteristic	GB/T 10220-2012	Characteristic
Taste	Intense sweetness	GB/T 10220-2012	Intense sweetness
Total protein (w/w%)	≥90%	AOAC 930.29	91.8%
Brazzein (w/w%)	≥90%	HPLC (QC-00-01)	91.0%
Loss on Drying (w/w%)	≤10%	AOAC 930.04	5.0%
Fat (w/w%)	≤1%	AOAC 996.06	ND
Ash (w/w%)	≤5%	AOAC 930.30	1.2%
Carbohydrates (w/w%)	≤ 5%	FDA 21 CFR 101.9	2.0%
Arsenic	≤0.2 ppm	AOAC 2015.01 ICP-MS	ND
Lead	≤0.1 ppm	AOAC 2015.01 ICP-MS	ND
Cadmium	≤0.1 ppm	AOAC 2015.01 ICP-MS	ND
Mercury	≤0.1 ppm	AOAC 2015.01 ICP-MS	ND
Total Plate Count	≤5,000 CFU/g	AOAC 990.12	550 CFU/g
Total Coliforms	≤10 CFU/g	AOAC 991.14	<10 CFU/g
Yeast & Molds	≤100 CFU/g	AOAC 997.02	<10 CFU/g
E. Coli	≤10 CFU/g	AOAC 991.14	<10 CFU/g
Salmonella	Not detected in 25g	AOAC 967.26	ND
Listeria	Not detected in 25g	AOAC 993.12	ND
Conclusion	The product complies with specification		
Storage	Store in a dry, room temperature environment; Keep away from direct sunlight; Ensure the storage area is free from strong odors.		

QC: [redacted]
Date: 2024.08.07

QA: [redacted]
Date: 2024.08.07

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水滴农厂 Farmtory

CERTIFICATION OF ANALYSIS			
Product Name	Farmtory Brazzein	Batch No.	20240820
Date of Manufacture	Aug.20,2024	Quantity	1000g
Date of Certificate	Aug.27,2024	Best Before Date	Feb.20,2026
Test	**Specification**	**Method**	**Result**
Appearance	Yellowish brown powder	GB/T 10220-2012	Yellowish brown powder
Odor	Characteristic	GB/T 10220-2012	Characteristic
Taste	Intense sweetness	GB/T 10220-2012	Intense sweetness
Total protein (w/w%)	$\geq$90%	AOAC 930.29	92.5%
Brazzein (w/w%)	$\geq$90%	HPLC (QC-00-01)	91.5%
Loss on Drying (w/w%)	$\leq$10%	AOAC 930.04	5.2%
Fat (w/w%)	$\leq$1%	AOAC 996.06	ND
Ash (w/w%)	$\leq$5%	AOAC 930.30	1.5%
Carbohydrates (w/w%)	$\leq$ 5%	FDA 21 CFR 101.9	0.8%
Arsenic	$\leq$0.2 ppm	AOAC 2015.01 ICP-MS	ND
Lead	$\leq$0.1 ppm	AOAC 2015.01 ICP-MS	ND
Cadmium	$\leq$0.1 ppm	AOAC 2015.01 ICP-MS	ND
Mercury	$\leq$0.1 ppm	AOAC 2015.01 ICP-MS	ND
Total Plate Count	$\leq$5,000 CFU/g	AOAC 990.12	720 CFU/g
Total Coliforms	$\leq$10 CFU/g	AOAC 991.14	<10 CFU/g
Yeast & Molds	$\leq$100 CFU/g	AOAC 997.02	<10 CFU/g
E. Coli	$\leq$10 CFU/g	AOAC 991.14	<10 CFU/g
Salmonella	Not detected in 25g	AOAC 967.26	ND
Listeria	Not detected in 25g	AOAC 993.12	ND
Conclusion	The product complies with specification		
Storage	Store in a dry, room temperature environment; Keep away from direct sunlight; Ensure the storage area is free from strong odors.		

QC:

Date: 2024.08.27

QA:

Date: 2024.08.27

3205940609334

Address: Suzhou Aquafarmtory Biotechnology LTD 68 Xinqing Road, Suzhou, Jiangsu,PRC,China
Contact: webpr@aquafarmtory.com
Website: www.aquafarmtory.com

Farmtory

CERTIFICATION OF ANALYSIS			
Product Name	Farmtory Brazzein	Batch No.	20240929
Date of Manufacture	Sept.29,2024	Quantity	1000g
Date of Certificate	Oct.21,2024	Best Before Data	Mar.29,2026
Test	**Specification**	**Method**	**Result**
Appearance	Yellowish brown powder	GB/T 10220-2012	Yellowish brown powder
Odor	Characteristic	GB/T 10220-2012	Characteristic
Taste	Intense sweetness	GB/T 10220-2012	Intense sweetness
Total protein (w/w%)	≥90%	AOAC 930.29	91.3%
Brazzein (w/w%)	≥90%	HPLC (QC-00-01)	90%
Loss on Drying (w/w%)	≤10%	AOAC 930.04	5.4%
Fat (w/w%)	≤1%	AOAC 996.06	ND
Ash (w/w%)	≤5%	AOAC 930.30	0.82%
Carbohydrates (w/w%)	≤5%	FDA 21 CFR 101.9	2.48%
Arsenic	≤0.2 ppm	AOAC 2015.01 ICP-MS	ND
Lead	≤0.1 ppm	AOAC 2015.01 ICP-MS	ND
Cadmium	≤0.1 ppm	AOAC 2015.01 ICP-MS	ND
Mercury	≤0.1 ppm	AOAC 2015.01 ICP-MS	ND
Total Plate Count	≤5,000 CFU/g	AOAC 990.12	640 CFU/g
Total Coliforms	≤10 CFU/g	AOAC 991.14	<10 CFU/g
Yeast & Molds	≤100 CFU/g	AOAC 997.02	<10 CFU/g
E. Coli	≤10 CFU/g	AOAC 991.14	<10 CFU/g
Salmonella	Not detected in 25g	AOAC 967.26	ND
Listeria	Not detected in 25g	AOAC 993.12	ND
Conclusion	The product complies with specification		
Storage	Store in a dry, room temperature environment; Keep away from direct sunlight; Ensure the storage area is free from strong odors.		

QC: [redacted]
Date: 2024.10.21

QA: [redacted]
Date: 2024.10.21

生物科技有
3205940609334

Address: Suzhou Aquafarmtory Biotechnology LTD 68 Xinqing Road, Suzhou, Jiangsu,PRC,China
Contact: webpr@aquafarmtory.com
Website: www.aquafarmtory.com

中国认可
国际互认
检测
TESTING
CNAS L4305

Test Report

NBF24-0018038-01 **Date:** 2024-10-21

Client Name: AQUAFARMTORY

Client Address: Building1,No.68XinqingRoad,Suzhou industrial Park,China(Jiangsu)pilot free trade zone

Sample Name:	AF-3
Sample Batch No.:	20240929
Production Date:	20240929
Manufacturer:	AQUAFARMTORY
Sample other information:	Main components:Protein Powder

Above information and sample(s) was/were submitted and certified by the client, SGS quoted the information with no responsibility as to the accuracy, adequacy and/or completeness.

Date of Sample Received:	2024-10-11
Testing Period:	2024-10-11 ~ 2024-10-21
Test Requested:	Select test(s) as requested by the client.
Test Method(s):	Please refer to next page(s).
Test Result(s):	Please refer to next page(s).

SGS Approved Signatory

Scan to see the report
NBF24-0018038-01
Verification:
check.sgsonline.com.cn

No.1177 Lingyun Road, Ningbo Hi-Tech Zone, Ningbo, China 315040 t 400-691-0488 www.sgsgroup.com.cn
中国・浙江・宁波高新区凌云路1177号 邮编：315040 t 400-691-0488 sgs.china@sgs.com

Test Report

NBF24-0018038-01 **Date:** 2024-10-21

Sample Description:

Sample No.	SGS Sample ID	Description
1	NBF24-0018038-0001	sample in bag

Test Result(s):

Microbial Test(s)

Code	Test Item(s)	Unit(s)	Test Method(s)	Test Result(s) 1
1	Total Plate Count	CFU/g	AOAC 990.12	6.4×10^2
2	Coliforms	CFU/g	AOAC 991.14	<10
3	Escherichia coli	CFU/g	AOAC 991.14	<10
4	Mould	CFU/g	AOAC 997.02	<10
5	Yeast	CFU/g	AOAC 997.02	<10
6	Salmonella	/25 g	AOAC 967.26	ND
7	Listeria monocytogenes	/25 g	AOAC 993.12	ND

Chemical Test(s)

Code	Test Item(s)	Unit(s)	Test Method(s)	Test Result(s) 1	LOQ
8	*Moisture	g/100 g	With reference to AOAC 930.04	5.40	-
9	Protein	%	AOAC 930.29	91.3	-
10	*Total Fat(as TG)	g/100 g	AOAC 996.06	ND	0.01
11	Ash	g/100 g	AOAC 930.30	0.82	-
12	*Lead (Pb)	mg/kg	AOAC 2015.01 ICP-MS	ND	0.02
13	*Cadmium (Cd)	mg/kg	AOAC 2015.01 ICP-MS	ND	0.02
14	*Arsenic (As)	mg/kg	AOAC 2015.01 ICP-MS	ND	0.02
15	*Mercury (Hg)	mg/kg	AOAC 2015.01 ICP-MS	ND	0.01
16	**Phosphorus(P)	mg/kg	With reference to AOAC 984.27	28.83	1.00
17	**Calcium(Ca)	mg/kg	With reference to AOAC 984.27	32.2	1.00
18	**Copper(Cu)	mg/kg	With reference to AOAC 984.27	3.9	1.00
19	**Iron(Fe)	mg/kg	With reference to AOAC 984.27	17.8	1.00
20	**Potassium(K)	mg/kg	With reference to AOAC 984.27	2.4	1.00
21	**Magnesium(Mg)	mg/kg	With reference to AOAC 984.27	6.1	1.00
22	**Manganese(Mn)	mg/kg	With reference to AOAC 984.27	0.3	0.10
23	**Sodium(Na)	mg/kg	With reference to AOAC 984.27	4011.4	1.00

No.1177 Lingyun Road, Ningbo Hi-Tech Zone, Ningbo, China 315040 t 400-691-0488 www.sgsgroup.com.cn
中国・浙江・宁波高新区凌云路1177号 邮编: 315040 t 400-691-0488 sgs.china@sgs.com

Test Report

NBF24-0018038-01 **Date:** 2024-10-21

Code	Test Item(s)	Unit(s)	Test Method(s)	Test Result(s) 1	LOQ
24	**Zinc(Zn)	mg/kg	With reference to AOAC 984.27	ND	1.00

Remark:
Conversion factor of nitrogen to protein is 6.25.

Remark:
1.ND = Not Detected
2.LOQ = Limit of Quantitation
3.*Test items were carried out by SGS-CSTC Standards Technical Services (Shanghai) Co.,Ltd. laboratory(CNAS No.L0599), Except Moisture, other items were not included in the CNAS/CMA Accredited Schedule for our laboratory.
4.**Test items were carried out by SGS-CSTC Standards Technical Services (Qingdao) Co., Ltd. laboratory), were not included in the CNAS Accredited Schedule for testing laboratory. and were not included in the CNAS Accredited Schedule for our laboratory.

Attention:
Unless otherwise stated the results shown in this test report refer only to the items tested.This document cannot be used for improper publicity, without prior written approval of the SGS. The test report shall only be used for scientific research, teaching, internal quality control.

*** End***

SGS-CSTC Standards Technical Services Co., Ltd. Ningbo Branch Agricultural and Food Laboratory
No.1177 Lingyun Road, Ningbo Hi-Tech Zone, Ningbo, China 315040 t 400-691-0488 www.sgsgroup.com.cn
中国・浙江・宁波高新区凌云路1177号 邮编：315040 t 400-691-0488 sgs.china@sgs.com

水滴农厂
Farmtory

Standard Operating Procedure for Determination of Farmtory Brazzein Content (HPLC Method)

Responsible person	Name	Signature	Date
QC	Ting Li		2024.01.08
QA	Hetian Liu		2024.01.08

File Code: QC-00-01
Version Number: QC-00-02

Address: Suzhou Aquafarmtory Biotechnology LTD 68 Xinqing Road, Suzhou, Jiangsu, PRC.
Contact: webpr@aquafarmtory.com
Website: www.aquafarmtory.com

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Appendix 5

1 Purpose

Standardizing the operation process for determining the content of Farmtory Brazzein (HPLC method).

2 Scope

This procedure is applicable for the determination of Farmtory Brazzein content by HPLC method.

3 Basis

This method is based on published protocol as reported.

4 Definition

HPLC: High Performance Liquid Chromatography

5 Content

5.1 Principle

The Brazzein is a protein compound that has a characteristic absorption peak in the UV absorption spectrum at 220 nm. The sample dissolved in water, separated by liquid chromatography, and detected by an UV detector, with quantification by the external standard method.

5.2 Experimental materials

5.2.1 Reagents and materials

5.2.1.1 Water: The resistivity is not less than 18.2 MΩ.cm, and all solutions used in this method are prepared with ultra-pure water.

5.2.1.2 Mobile phase A 0.1% trifluoroacetic acid solution: Taking 1 mL of trifluoroacetic acid and adding it to 999 mL water, mix well, filter through a 0.22 μm filter membrane, and degassing by ultrasonic treatment for later use.

Mobile phase B 0.1% trifluoroacetic acid +100% acetonitrile: Taking 1 mL of trifluoroacetic acid and adding to 999 mL HPLC-grade acetonitrile, mix well, filter through a 0.22 μm filter membrane, and degassing by ultrasonic treatment for later use.

5.2.1.3 Brazzein standard: Brazzein standard from Nanjing Warbio Biotechnology Co Ltd, catalog number Wb2641, purity 98.8%.

Brazzein preliminary reference standard: Nanjing Warble Biotechnology, catalog number Wb2641 and secondary reference standard: qualified Farmtory Brazzein lot 20240102 with preliminary reference standard.

5.2.2 Instruments and equipment

High-performance liquid chromatography equipment (Thermo Scientific Vanquish), Precision analytical balance (METTLER TOLEDO AL104), Vortex oscillator, Volumetric flask, Syringe, 0.22 μm filter membrane, Sample vial.

5.2.3 Chromatographic analysis conditions

HPLC Instrument model：Thermo Scientific Vanquish Core

Chromatography Column：Luna® 5um C18(2) 100Å，250×4.6 mm

Mobile Phase:

Mobile Phase A: 0.1% trifluoroacetic acid

Mobile Phase B: 0.1% trifluoroacetic acid+100% acetonitrile

Detection Wavelength: UV220 nm

Column Temperature: 37 °C

Injection Volume: 10 μL

Flow Rate: 0.5 mL/min

Run Time: 15 min

Gradient Program:

Time	Flow(mL/min)	%A	%B
0.000	0.5	70	30
0.100	0.5	70	30
10.000	0.5	10	90
10.100	0.5	70	30
15.000	0.5	70	30

5.2.4 Analytical steps

5.2.4.1 Preparation of Brazzein standard solution

Standard solution preparation:

1. Accurately weigh 0.0100 g of the Brazzein standard to the nearest 0.0001 g.
2. Dissolve the weighed Brazzein in a suitable amount of water.
3. Transfer the solution to a 10 mL volumetric flask.
4. Rinse the container used for weighing with a small amount of water and transfer the rinse to the volumetric flask.
5. Dilute the solution to the mark with water and mix well. This will result in a stock solution of 1 g/L (1000 mg/L) Brazzein.

Standard working solution preparation:

Dilute the standard stock solution (prepared previously at 1 g/L) to prepare a series of mixed standard solutions with concentrations of 0.1 g/L, 0.2 g/L, 0.5 g/L, 0.75 g/L, and 1 g/L.

5.2.4.2 Preparation of Farmtory Brazzein sample solution

1. Accurately weigh 0.1000 g of the Farmtory Brazzein sample to the nearest 0.0001 g.
2. Dissolve the weighed sample in a suitable amount of water in a clean container.
3. Transfer the dissolved sample to a 100 mL volumetric flask.
4. Rinse the container used for weighing with a small amount of water and transfer the rinse to the volumetric flask.
5. Dilute the solution to the mark with water and mix well to ensure complete dissolution.
6. If necessary, further dilute the resulting solution to bring the concentration within the linear range of the analytical curve (based on the actual concentration determined previously or estimated).

7. A total of two Farmtory Brazzein sample solutions were prepared as described above, and each was injected at least twice.

5.2.4.3 Determination

5.2.4.3.1 Connect the mobile phase and flush the chromatography system.

First, rinse the tubing with pure water to displace the organic phase in the tubing. Then, displace the pure water in the tubing with the mobile phase.

5.2.4.3.2 Column equilibration

After connecting the C18 chromatography column, equilibrate the column with a mixture of 70% mobile phase A and 30% mobile phase B at a flow rate of 0.5 mL/min until a stable and flat baseline is achieved.

5.2.4.3.3 System suitability testing

After the chromatographic parameters are stable, the prepared standard stock solution (1 g/L) is filtered through a 0.22 μm filter membrane, then transferred into dedicated sample bottle. The bottle is placed in the sample tray, and the corresponding position number is entered into the sequence table. Next, 10μL of each sample is loaded, and the sequence is run. Testing at least six times for system suitability.

5.2.4.3.4 Sample measurement

The prepared standard solutions and sample solutions are filtered through a 0.22 μm filter membrane and then transferred into dedicated sample bottles. These bottles are placed in the sample tray, and the corresponding position numbers are entered into the sequence table. Next, 10μL of each sample is loaded, and the sequence is run. After the detection is completed, the peak area and retention time of the main peak (target protein peak) are obtained based on the integration parameters set in the chromatography software. Using the external standard method, a standard curve is created using a series of standard solutions. The reference retention time and chromatogram for Brazzein are provided in Appendix A.

5.2.5 Result calculation

5.2.5.1 System suitability calculation

Organizing the statistical data of Brazzein standard stock solution of six times system suitability testing, and the RSD value of the target protein peak area is less than or equal to 1%.

The standard curve was fitted by Sigma Plot software to obtain the standard curve equation, with an R^2 value above 0.999.

5.2.5.2 Farmtory Brazzein content calculation

The content of Farmtory Brazzein (W) is calculated according to formula (A.1).

$$W = \frac{c \times V}{m \times 1000} \times f \times 100\% \quad \text{(A.1)}$$

In the formula:

c: The concentration of Farmtory Brazzein in the sample solution, in units of g/L, was calculated from the Brazzein standard curve

V: Volume of the test solution, in units of mL

m: Sample mass, in units of g

f: Dilution factor

The test results shall be based on the arithmetic mean of the results of parallel determinations. The RSD value of Farmtory Brazzein content (W) of the difference between two independent determinations is less than or equal to 1%.

6 Reference

1. Fariba M. Assadi-porter, David J. Aceti, John L. Markly (2000). Sweetness Determinant Sites of Brazzein, a Small, Heat-Stable, Sweet-Tasting protein. Archives of Biochemistry and Biophysics. Vol. 376, No. 2, April 15, pp.259-265, doi:10.1006/abbi.2000.1726.

Appendix A

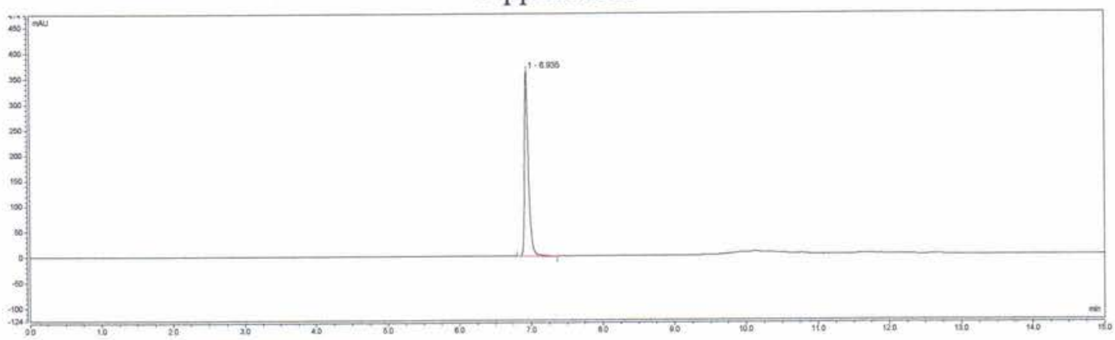

Figure A1. HPLC chromatogram of Brazzein standard

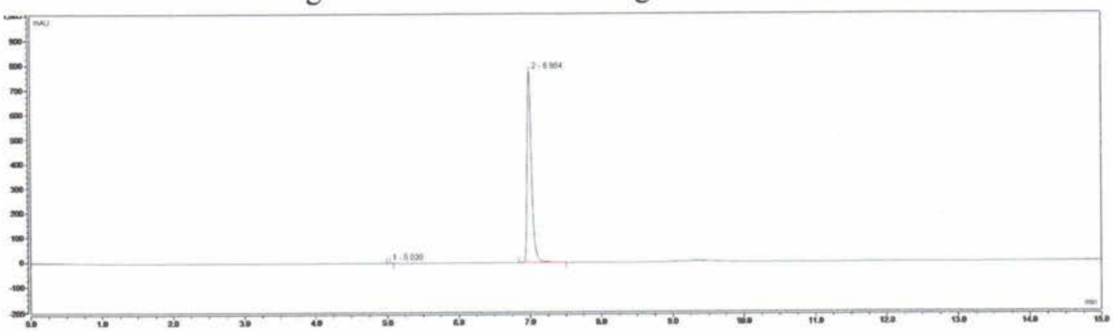

Figure A2. HPLC chromatogram of Farmtory Brazzein

Note：HPLC=High Performance Liquid Chromatography

Form Approved: OMB No. 0910-0342; Expiration Date: 08/31/2025
(See last page for OMB Statement)

FDA USE ONLY

GRN NUMBER	DATE OF RECEIPT
ESTIMATED DAILY INTAKE	INTENDED USE FOR INTERNET
NAME FOR INTERNET	
KEYWORDS	

DEPARTMENT OF HEALTH AND HUMAN SERVICES
Food and Drug Administration

GENERALLY RECOGNIZED AS SAFE *(GRAS)* NOTICE (Subpart E of Part 170)

Transmit completed form and attachments electronically via the Electronic Submission Gateway *(see Instructions)*; OR Transmit completed form and attachments in paper format or on physical media to: Office of Food Additive Safety *(HFS-200)*, Center for Food Safety and Applied Nutrition, Food and Drug Administration,5001 Campus Drive, College Park, MD 20740-3835.

SECTION A – INTRODUCTORY INFORMATION ABOUT THE SUBMISSION

1. Type of Submission *(Check one)*

☒ New ☐ Amendment to GRN No.________ ☐ Supplement to GRN No. ________

2. ☒ All electronic files included in this submission have been checked and found to be virus free. *(Check box to verify)*

3 Most recent presubmission meeting *(if any)* with FDA on the subject substance *(yyyy/mm/dd)*: ________

4 For Amendments or Supplements: Is your amendment or supplement submitted in response to a communication from FDA? *(Check one)* ☐ Yes ☐ No If yes, enter the date of communication *(yyyy/mm/dd)*: ________

SECTION B – INFORMATION ABOUT THE NOTIFIER

1a. Notifier

Name of Contact Person	Position or Title
Oliver Yu, Ph.D.	Chief Science Officer

Organization *(if applicable)*: MicroFarmtory LLC

Mailing Address *(number and street)*: 163 Capitol Ave, Suite 413, C1088,

City	State or Province	Zip Code/Postal Code	Country
Cheyenne	Wyoming	82001	United States of America

Telephone Number	Fax Number	E-Mail Address
314-753-0042		oliver.yu@microfarmtory.com

1b. Agent or Attorney *(if applicable)*

Name of Contact Person	Position or Title
Wenwen Ma, Ph.D.	Regulatory specialist

Organization *(if applicable)*: PureNutrix+ Inc.

Mailing Address *(number and street)*: 400 West Capitol Avenue, Suite 1700

City	State or Province	Zip Code/Postal Code	Country
Little Rock	Arkansas	72201	United States of America

Telephone Number	Fax Number	E-Mail Address
501-800-4108		regulatory@purenutrix.us

SECTION C – GENERAL ADMINISTRATIVE INFORMATION

1. Name of notified substance, using an appropriately descriptive term

GRAS NOTICE FOR A SWEET PROTEIN FARMTORYTM BRAZZEIN

2. Submission Format: *(Check appropriate box(es))*

☒ Electronic Submission Gateway

☐ Electronic files on physical media

☐ Paper

If applicable give number and type of physical media

3. For paper submissions only:

Number of volumes

Total number of pages

4. Does this submission incorporate any information in CFSAN's files? *(Check one)*

☐ Yes *(Proceed to Item 5)* ☒ No *(Proceed to Item 6)*

5. The submission incorporates information from a previous submission to FDA as indicated below *(Check all that apply)*

☐ a) GRAS Notice No. GRN

☐ b) GRAS Affirmation Petition No. GRP

☐ c) Food Additive Petition No. FAP

☐ d) Food Master File No. FMF

☐ e) Other or Additional *(describe or enter information as above)*

6. Statutory basis for conclusions of GRAS status *(Check one)*

☒ Scientific procedures *(21 CFR 170.30(a) and (b))* ☐ Experience based on common use in food *(21 CFR 170.30(a) and (c))*

7. Does the submission (including information that you are incorporating) contain information that you view as trade secret or as confidential commercial or financial information? (see 21 CFR 170.225(c)(8))

☐ Yes *(Proceed to Item 8*

☒ No *(Proceed to Section D)*

8. Have you designated information in your submission that you view as trade secret or as confidential commercial or financial information *(Check all that apply)*

☐ Yes, information is designated at the place where it occurs in the submission

☐ No

9. Have you attached a redacted copy of some or all of the submission? *(Check one)*

☐ Yes, a redacted copy of the complete submission

☐ Yes, a redacted copy of part(s) of the submission

☐ No

SECTION D – INTENDED USE

1. Describe the intended conditions of use of the notified substance, including the foods in which the substance will be used, the levels of use in such foods, and the purposes for which the substance will be used, including, when appropriate, a description of a subpopulation expected to consume the notified substance.

The intended use of FarmtoryTM Brazzein as a general-purpose sweetener in food, excluding infant formula, meat, and poultry products

2. Does the intended use of the notified substance include any use in product(s) subject to regulation by the Food Safety and Inspection Service (FSIS) of the U.S. Department of Agriculture?

(Check one)

☐ Yes ☒ No

3. If your submission contains trade secrets, do you authorize FDA to provide this information to the Food Safety and Inspection Service of the U.S. Department of Agriculture?

(Check one)

☐ Yes ☐ No , you ask us to exclude trade secrets from the information FDA will send to FSIS.

SECTION E – PARTS 2 -7 OF YOUR GRAS NOTICE

(check list to help ensure your submission is complete – PART 1 is addressed in other sections of this form)

☒ PART 2 of a GRAS notice: Identity, method of manufacture, specifications, and physical or technical effect (170.230).

☒ PART 3 of a GRAS notice: Dietary exposure (170.235).

☒ PART 4 of a GRAS notice: Self-limiting levels of use (170.240).

☒ PART 5 of a GRAS notice: Experience based on common use in foods before 1958 (170.245).

☒ PART 6 of a GRAS notice: Narrative (170.250).

☒ PART 7 of a GRAS notice: List of supporting data and information in your GRAS notice (170.255)

Other Information

Did you include any other information that you want FDA to consider in evaluating your GRAS notice?

☐ Yes ☒ No

Did you include this other information in the list of attachments?

☐ Yes ☐ No

SECTION F – SIGNATURE AND CERTIFICATION STATEMENTS

1. The undersigned is informing FDA that MicroFarmtory LLC

(name of notifier)

has concluded that the intended use*(s)* of FarmtoryTM Brazzein

(name of notified substance)

described on this form, as discussed in the attached notice, is (are) not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act based on your conclusion that the substance is generally recognized as safe recognized as safe under the conditions of its intended use in accordance with § 170.30.

2. MicroFarmtory LLC agrees to make the data and information that are the basis for the conclusion of GRAS status available to FDA if FDA asks to see them;

(name of notifier)

agrees to allow FDA to review and copy these data and information during customary business hours at the following location if FDA asks to do so; agrees to send these data and information to FDA if FDA asks to do so.

Wenwen Ma, Ph.D., PureNutrix+ Inc, 400 West Capitol Avenue, Suite 1700, Little Rock, AR, 72201, Tel: 501-800-4108; Email: re

(address of notifier or other location)

The notifying party certifies that this GRAS notice is a complete, representative, and balanced submission that includes unfavorable, as well as favorable information, pertinent to the evaluation of the safety and GRAS status of the use of the substance.The notifying party certifies that the information provided herein is accurate and complete to the best or his/her knowledge. Any knowing and willful misinterpretation is subject to criminal penalty pursuant to 18 U.S.C. 1001.

3. Signature of Responsible Official, Agent, or Attorney	Printed Name and Title	Date (mm/dd/yyyy)
Wma Digitally signed by Wenwen Ma Date: 2025.07.29 20:22:30 +02'00'	Wenwen Ma, Ph.D., Regulatory Specialist	07/17/2025

SECTION G – LIST OF ATTACHMENTS

List your attached files or documents containing your submission, forms, amendments or supplements, and other pertinent information. Clearly identify the attachment with appropriate descriptive file names (or titles for paper documents), preferably as suggested in the guidance associated with this form. Number your attachments consecutively. When submitting paper documents, enter the inclusive page numbers of each portion of the document below.

Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	Form3667.pdf	Administrative
	FarmtoryTMBrazzeinGRASNotification2025-07-26.pdf	GRAS Notice
	COSM_3667_26247_MicroFarmtoryLLC.pdf	Administrative
	SignedCoverletterbrazzein250725.pdf	Administrative